



KALASALINGAM
ACADEMY OF RESEARCH AND EDUCATION
(DEEMED TO BE UNIVERSITY)
Under sec. 3 of UGC Act 1956. Accredited by NAAC with "A++" Grade



SCHOOL OF ELECTRONICS, ELECTRICAL AND BIOMEDICAL TECHNOLOGY
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

PRESENTS



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ABOUT THE DEPARTMENT

The Department of Electronics and Communication Engineering (ECE) at Kalasalingam Academy of Research and Education (KARE) is pioneer in academic excellence and innovative research. Accredited by ABET (Accreditation Board for Engineering and Technology, USA) and NBA (National Board of Accreditation, India), the department adheres to global standards in education and research.

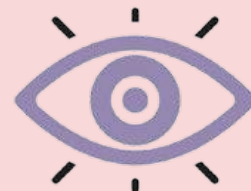
The department has been strengthened by support from prestigious initiatives such as:

- DST-FIST (Department of Science and Technology - Fund for Improvement of S&T Infrastructure), enhancing state-of-the-art laboratories and fostering advanced research.
- AICTE MODROB funding, which has facilitated the modernization of the Embedded Systems and Robotics Laboratory, equipping it with cutting-edge tools for innovation in microcontrollers and robotics.
- Through collaborations with industry partners, the department offers industry-supported courses designed to align academic learning with real-world applications. Notable partnerships include Nanochip Technologies, P2F Semi, and MathWorks, enabling students to gain expertise in VLSI, IoT, embedded systems, and sustainable technologies relevant to industrial demands.
- The department's focus on experiential learning includes hands-on projects, industry internships, and problem-based learning, ensuring graduates are well-equipped to address global engineering challenges and contribute to sustainable development.

Department of Electronics and Communication Engineering

INSTITUTE VISION:

To be a University of Excellence of International Repute in Education and Research.



INSTITUTE MISSION:



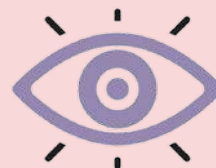
To provide a scholarly teaching-learning ambience which results in creating graduates equipped with skills and acumen to solve real-life problems.

To promote research and create knowledge for human welfare, rural and societal development.

To nurture entrepreneurial ambition, industrial and societal connect by creating an environment through which innovators and leaders emerge.

DEPARTMENT VISION:

To strive towards excellence in Electronics and Communication engineering through teaching, experiential learning, quality research and scholarship, while adhering to ethical and societal requirements



DEPARTMENT MISSION:

To help students achieve their goals by recognizing, identifying, and fostering their unique strengths through quality education and cutting-edge research training.

To imbibe ability in the students to solve real life problems as per need of the society through nurturing their skills, creative thinking, and research acumen.

To create an elite workforce, without compromise in the ethics and societal values.

Program Educational Objectives (PEOs) (After June 2021)

PEO1: Technical Proficiency

Succeed as creative, productive, and valued engineers in their career by applying the technical knowledge and skills gained.

PEO2: Professional Growth

Continue to develop professionally through life-long learning, higher education, and research expertise to meet the expectations of their organisation and the society.

PEO3: Management Skills

Exhibit the management qualities in a responsive, ethical, and innovative manner.

B.TECH ECE PROGRAMME SPECIFIC OUTCOMES

At the end of the programme, the students will be able to:

PSO1: Knowledge:

Apply the knowledge of engineering and basic sciences to design, develop, test components and systems for applications including Signal Processing, VLSI Design, Embedded, Communication Engineering and Networking

PSO2: Skills:

Solve the complex engineering problems, using latest techniques, tools, along with the needed skills with an understanding of societal, environmental, safety, legal and cultural impacts of the solution.

PSO3: Attitude:

Apply the contextual knowledge of engineering to function effectively as an individual or a leader in multidisciplinary environments

B.TECH ECE PROGRAMME OUTCOMES (R2021)

At the end of the programme, the students will be able to:

PO1: Engineering knowledge:

Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem analysis:

Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/development of solutions:

Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct investigations of complex problems:

Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering tool usage:

Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World:

Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

B.TECH ECE PROGRAMME OUTCOMES (R2021)

PO7: Ethics:

Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work:

Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication:

Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

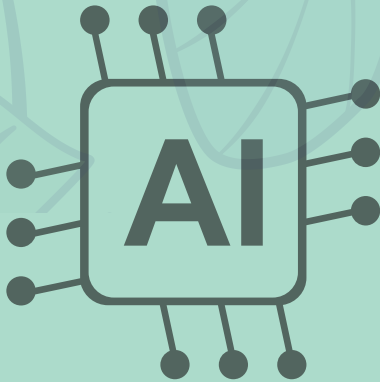
PO10: Project Management and Finance:

Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning:

Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

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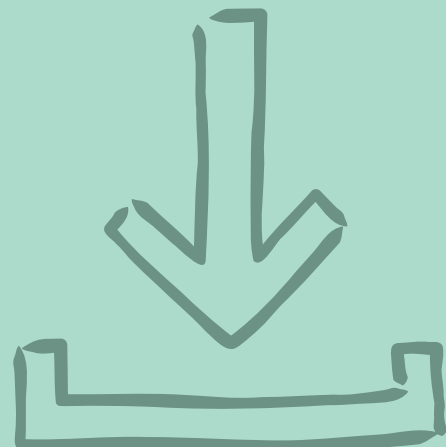
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Dear Readers

Welcome to the latest issue of ELECTROCOMM -Volume 16, Issue 32. It is with great pleasure and enthusiasm that we present to you a diverse collection of stories, insights, and inspiration curated for your reading pleasure.

In the fast-paced world we live in ELECTROCOMM aims to be your guide, offering a moment of respite and discovery.

As we flip through these pages together, you will encounter thought-provoking articles, captivating visuals, and interviews with individuals who are making waves in their respective fields. Our goal is to create a space for meaningful dialogue, celebration of accomplishments, and exploration of the ideas shaping our world.

A magazine is more than just a collection of words and images; it's a journey. It's an exploration of new concepts, a celebration of diversity, and a platform for voices that deserve to be heard.

We extend our deepest gratitude to our talented contributors, whose creativity and dedication have brought this magazine to life. We also want to thank you, our readers, for your continued support and for being an essential part of the ELECTROCOMM community.

As you delve into the pages ahead, we encourage you to immerse yourself fully in the stories, connect with the ideas, and share your thoughts with us. Your feedback is invaluable and helps us grow.

Dr. J. Charles Pravin
HOD, ECE



Warm Greetings to all!!!

It gives me immense joy and satisfaction to pen in the 32nd Issue, of our Department Magazine 'ELECTROCOMM' on this occasion as Editor. Magazines are the perfect platform that develops one's creativity. The biggest blocks to creativity is the inability to allow a thought or insight to have the space to become a possibility. Some of the inspired idea are the ones that start as crazy concepts. By allowing them to possibility, they may blossom into something usable. The Magazine aims to encourage creativity among young minds and inspire self-motivation. This issue includes articles, poems, sketches, photographs and technical, non-technical writeups and stories.

In each one, the contributors demonstrated their best aspects of their personalities through self-expression.

I would like to thank each and every contributor for their active participation in making the Magazine to float in our hands.

As an Editor, I hope you too enjoy reading this Magazine as much as we have enjoyed creating it.

A handwritten signature in black ink, which appears to read 'Abha'.

Dr. Abha Dargar
Asst. Prof, ECE

◀ MEET THE TEAM ▶

★ Convener

Dr. J. Charles Pravin
HOD, ECE

★ Faculty Co-Ordinator

Dr. Abha Dargar
Asst. Prof / ECE

★ Student Coordinators

Avulapati Giridhar
P. Vamsi Krishna
Sai Siddarth S

★ Design Team

Dinesh Naidu
Shanmukha Srihari
Ajay Kumar
Aakash Bala Prajapathi
Neelakanteswar Reddy
Shaik Jafiya Naz

ELECTROCOMM
MAGAZINE

NAMBI NARAYANAN

FALSE CHARGES. ABSOLUTE TRUTH..



NAMBI NARAYANAN

A Living Blueprint of Rocket Science in Action

Nambi Narayanan was the architect of the Liquid Propulsion Systems in India. He was instrumental in the development of the Vikas engine, the workhorse liquid fuel engine that powers ISRO's PSLV and GSLV rockets. His team's efforts involved reverse-engineering and innovating upon technology from France, laying the foundation for India's autonomous satellite launch capabilities.

His work underscored the importance of precision, thermal design, and materials science in extreme environments. The technology and systems he pioneered not only enabled successful satellite launches but also formed the critical base for India's future interplanetary missions, including the Chandrayaan and Mangalyaan missions. His strategic thinking charted the course for India's ascent as a global space power

A Legacy Honored: The Triumph of Truth

Narayanan is a hero who was first accused, then celebrated. The Supreme Court, in a landmark judgment, acknowledged him as a victim of a fabricated case, highlighting the "moral harassment" he endured. He has been awarded the Padma Bhushan, India's third-highest civilian award, and his story has been immortalized in the feature film "Rocketry: The Nambi Effect,"

ORIGINS OF A VISIONARY

Nambi Narayanan was born on December 12, 1941, in Nagercoil, in the erstwhile Travancore state (present-day Tamil Nadu). From a young age, he exhibited a profound fascination for engineering and science. He earned a degree in Mechanical Engineering from the University of Madras and later secured a scholarship to pursue a master's degree in Aerospace Engineering at the prestigious Princeton University in the United States. This move would set the stage for his groundbreaking future.

FROM THE CLASSROOMS OF PRINCETON TO THE INDIAN SPACE PROGRAM:

Upon returning to India, driven by a powerful vision of self-reliance, Narayanan joined the Indian Space Research Organisation (ISRO) in 1966. He quickly became a pivotal figure in the fledgling space program, working under the legendary Dr. Vikram Sarabhai. His work at Princeton and his innate genius made him a natural choice to lead the development of liquid fuel rocket engine technology in India.

The keypoint of his legacy is not just his scientific genius, but his unwavering integrity. As a result of his ordeal, the nation was forced to confront the perils of false allegations and bureaucratic malfeasance. His greatest triumph was his vindication, which restored his honor and optimised the spirit of scientific pursuit in India.





Faculty Articles

Silicon Photonics: Lighting the Way to the Future

If you've ever wished your internet could move faster than your late-night exam panic, silicon photonics might just be the genie you were looking for. Simply put, silicon photonics is the clever fusion of silicon microelectronics with photonic circuits—the marriage of chips and light. Instead of pushing sluggish electrons around, we're now zipping information with photons. The result? Blazing-fast data transfer, sky-high bandwidth, and serious energy savings.

In just over a decade, this field has grown from lab experiments to millions of devices in action, mostly as optical transceivers powering the world's data centers. But hold on—this isn't just about faster Netflix streaming or quicker downloads. Silicon photonics is already spreading its wings into sensing, computing, and healthcare, making it one of the most exciting playgrounds for engineers and researchers today.

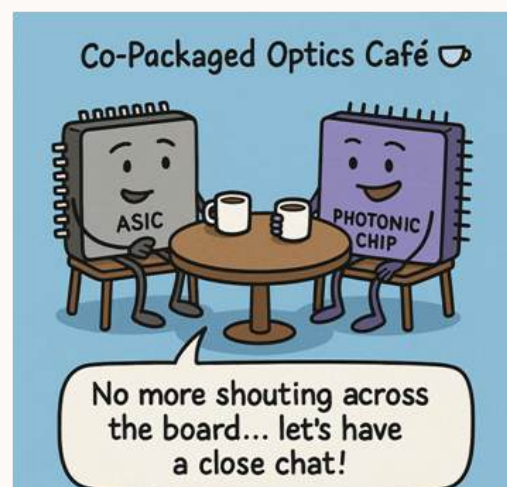
Breakthroughs and Adoption:

Modern photonic integrated circuits (PICs) are like the Swiss Army knives of light—they can pack lasers, modulators, waveguides, and detectors all onto one tiny silicon chip. Of course, silicon itself isn't the best at generating laser light (it's a bit shy because of its indirect bandgap). Engineers, never short on creativity, solved this by integrating III-V materials like indium phosphide, giving silicon a little “laser buddy.”

Modulators have also had a glow-up, moving from plain old plasma dispersion effects to glamorous materials like thin-film lithium niobate, enabling ultra-fast, low-power modulation. Meanwhile, photodetectors are now

made with SiGe (silicon-germanium), which happily fits into standard CMOS processes. Translation: cheaper, faster, and easier to manufacture.

The packaging story is equally exciting. Remember those big pluggable optical modules? They're being replaced by co-packaged optics (CPO), where optics and electronics sit shoulder-to-shoulder, practically whispering to each other.



This shrinks electrical paths from centimeters to millimeters—saving power, reducing latency, and upping bandwidth. Basically, it’s like moving from shouting across a football field to having a quiet coffee chat with your ASIC. It’s no surprise then that big players like Intel and Cisco, along with startups galore, are betting on silicon photonics. With initiatives like the U.S. National Photonics Initiative and AIM Photonics institute, the field has shifted from an underdog challenger to the go-to technology for interconnects.

India’s Emerging Ecosystem:

While the US, Europe, and East Asia had a head start, India is now catching up with serious momentum. The Centre for Programmable Photonic Integrated Circuits and Systems (CPPICS) at IIT Madras is a shining example, launched in 2023 with MeitY’s support. Through collaborations with Si2 Microsystems (for packaging) and SilTerra (for fabrication), CPPICS has put India on the silicon photonics map.

The results are already visible: a fiber-array attachment tool (think of it as precision Lego for aligning fibers to chips) and a quantum random number generator (QRNG) module for DRDO—because why settle for pseudo-random when you can have truly quantum-random? These projects aren’t just research trophies; they’re stepping stones toward India’s self-reliance in high-tech industries.

Transformative Applications

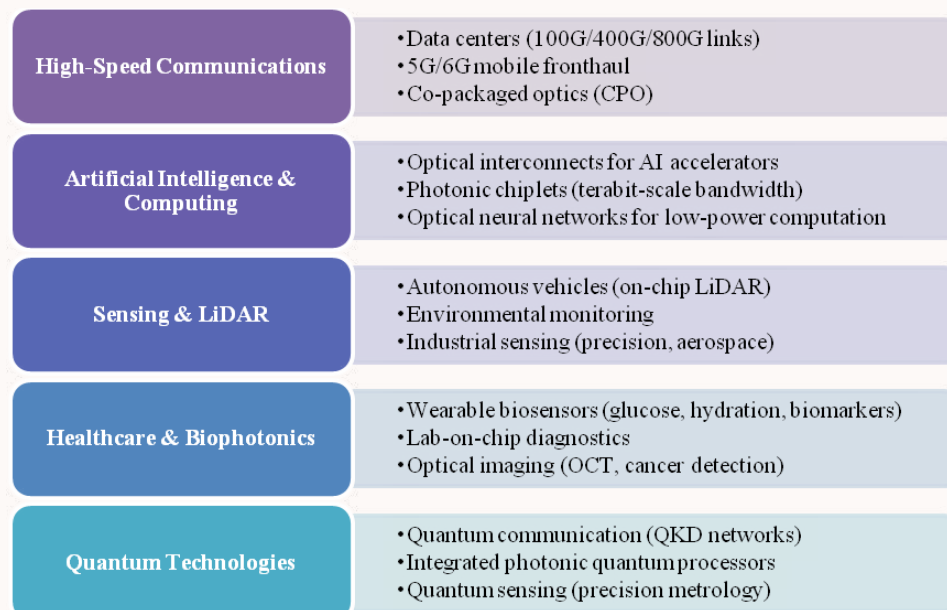


Figure: Scope of Silicon Photonics showing its transformative impact on next-generation systems and applications.

High-Speed Communications – Silicon photonics is the backbone of today’s data centers and is gearing up for 5G/6G and terabit switches. With CPO in play, the dream of faster and cooler networks is no longer science fiction.

Artificial Intelligence & Computing – Photonic chiplets are here to rescue AI from its data bottlenecks. Imagine moving terabits of data per second at light speed. Bonus: optical neural networks might one day let us run AI without guzzling megawatts of power.

Sensing & LiDAR – Autonomous vehicles need eyes, and silicon photonic LiDAR is giving them vision. Add in spectrometers for environmental monitoring and sensors for aerospace, and suddenly photonics is everywhere—from cars to climate labs.

Healthcare & Biophotonics – Fancy a smartwatch that checks your glucose, hydration, or even lactate levels in real time? Silicon photonic biosensors can make that happen. Lab-on-chip diagnostics and portable imaging will soon make a doctor’s visit as simple as strapping on a wearable.

Quantum Technologies – Whether it’s secure communication, quantum processors, or ultra-precise sensors, silicon photonics is the quiet hero enabling scalable quantum tech. It’s like giving quantum physics a silicon passport to the real world.

Careers and Skills for ECE Students

Here’s the best part for students: silicon photonics isn’t just about cool science; it’s also about hot careers. From PIC design and fabrication to system integration and testing, the industry needs bright minds who can juggle optics, electronics, and software.

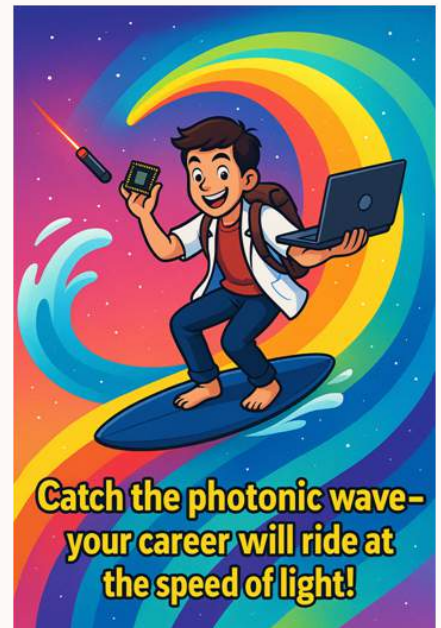
To get ahead, you’ll want a toolbox with:

- Optics & Photonics fundamentals (waveguides, lasers, comms systems),
- Electronics & RF know-how (drivers, amplifiers, SerDes),
- Semiconductor processing basics (CMOS, MEMS, materials), and
- Design software chops (Lumerical, Synopsys, MATLAB, Python).

Whether you aim for research labs, startups, or tech giants, this field is like a buffet of opportunities. Sure, advanced degrees help, but hands-on projects, internships, and photonics hackathons can be your golden ticket. And don’t forget—communication skills matter. You’ll often be explaining waveguide bends and laser bonding to teammates who may or may not think photons are magic.

In just two decades, silicon photonics has sprinted from a niche curiosity to a cornerstone of modern communication and computing. Its fingerprints are already on the cloud, AI, healthcare, and soon, quantum tech. For ECE students, it's more than just another subject—it's a career-defining frontier that blends optics, electronics, and imagination

So, dear students, don't just sit back and watch the photons race by—catch them, ride the wave, and let silicon photonics brighten your career path (at the speed of light, of course!).



Dr. Jenila C,
Assistant Professor, ECE

India Semiconductor Mission and the Emerging Fabless Design Ecosystem

Background:

The semiconductor industry forms the foundation of the modern digital economy, powering everything from smartphones and cars to satellites and AI systems. To strengthen India's position in this strategic domain, the Government of India launched the India Semiconductor Mission (ISM) under the Digital India initiative. The mission aims to develop a sustainable semiconductor and display manufacturing ecosystem within the country, encouraging fabrication units (fabs) as well as fabless design companies. Supported by policy incentives, global partnerships, and skill development programs, ISM envisions transforming India into a global semiconductor hub.

Introduction:

India's dependence on imported semiconductor chips accounting for nearly \$20 billion annually poses significant supply chain risks. To address this, the India Semiconductor Mission (ISM) was introduced in 2021 to support semiconductor manufacturing and design at all levels, from material sourcing to chip packaging. ISM's initiatives, combined with India's growing fabless design sector, aim to establish a complete semiconductor value chain spanning design, fabrication, testing, and deployment. India is also nurturing a new wave of fabless startups, backed by educational and R&D initiatives from IITs, IISc, and NITs. Together, they are bridging the gap between academic innovation and industrial application, positioning India as a global chip design powerhouse. The India Semiconductor Mission (ISM) functions under MeitY (Ministry of Electronics and Information Technology), focusing on:

- Silicon and Compound Semiconductor Fabs
- Display Fabs
- Assembly, Testing, Marking, and Packaging (ATMP/OSAT)
- Design-Linked Incentive (DLI) and CHIP to Startup (C2S) Programmes
- The CHIP to Startup (C2S) Programme

The "Chip to Startup (C2S)" program launched by MeitY, is a flagship initiative under ISM aimed at training 85,000 engineers in semiconductor design and innovation. It supports over 100 academic and R&D institutions across India to develop ASICs, System-on-Chip (SoC), and IP cores using advanced EDA tools and fabrication support. The C2S programme connects academia, startups, and industries, providing mentorship, funding, and access to multi-project wafer (MPW) fabrication facilities. This initiative ensures that India's talent base aligns with the global semiconductor industry's technical needs.

Recent Developments: Semiconductor Projects in India

New Chips Fall in Place for India's Semicon Mission

The India Semiconductor Mission (ISM) has approved ten semiconductor units who will all receive central and state government subsidies under the programme for development of semiconductors and display manufacturing ecosystem in India. The total approved projects' cumulative investments has reached around ₹1.6 lakh crore in six Indian States. ET provides a quick reckoner of the ten units along with those in the pipeline.

Micron Technology

June, 2023: Union Cabinet approves proposal of Micron Technology for setting up chip ATMP (assembly, testing, marking and packaging) unit in Sanand, Gujarat

Construction: First phase of Micron's Sanand facility undergoing clean room validation for partial operation in late 2025

• Memory and storage products manufactured at the ATMP unit will largely be export-led

Total proposed investment
\$2.75 billion

Micron's commitment
\$825 million

The rest: State and central government subsidies



Tata Electronics Private Limited (TEPL) and Powerchip Semiconductor Manufacturing Corp (PSMC)

• To set up semiconductor fabrication unit

Capacity:
50,000 wafer starts per month

Location: Dholera, Gujarat

Investment: ₹91,000 crores

To produce:
High performance 28 nm compute chips

To produce: Power management chips for electric vehicles, telecom, defence, automotive, consumer electronics, display, and power electronics

Tata Semiconductor Assembly and Test (TSAT)

• To set up ATMP unit in Morigaon, Assam

Investment:
₹27,000 crores

Capacity:
48m per day

To develop: Advanced semiconductor packaging technologies including flip chip and integrated system in package technologies

Segments: Automotive, electric vehicles, consumer electronics, telecom, and mobile phones

CG Power, Renesas Electronics Corporation, Japan and Stars Microelectronics, Thailand

• To set up ATMP unit

Investment:
₹7,600 crores

Capacity:
15 million per day

Location: Sanand, Gujarat

• Unit to manufacture chips for consumer, industrial, automotive and power applications

Under ISM, India has approved multiple semiconductor units with central and state government support, amounting to ₹1.6 lakh crore in cumulative investments across six Indian states (Source: Economic Times, June 2023).

Micron Technology – Sanand, Gujarat:

- Investment: \$2.75 billion (~₹22,000 crore)
- Focus: Memory and storage products (largely export-led)
- Status: Clean room validation under progress; partial operation by 2025.

Tata Electronics Pvt. Ltd (TEPL) & Powerchip Semiconductor Manufacturing Corp (PSMC) – Dholera, Gujarat

- Investment: ₹91,000 crore
- Capacity: 50,000 wafer starts per month
- Output: Power management and high-performance 28 nm compute chips for EVs, defence, and telecom.

Tata Semiconductor Assembly and Test (TSAT) – Morigaon, Assam

- Investment: ₹27,000 crore
- Capacity: 48 million packages per day
- Focus: Advanced semiconductor packaging technologies.

These projects mark India's transition from a design-focused economy to an integrated semiconductor powerhouse.

Fabless Design Ecosystem in India:

The fabless model focuses on chip design, verification, and IP development, while fabrication is outsourced to foundries abroad. India's strength lies in its skilled design workforce nearly 20% of the world's semiconductor design engineers are based in India. Leading global companies such as Qualcomm, Intel, AMD, and NVIDIA operate major design centers in Bengaluru, Hyderabad, and Noida, while indigenous startups like Saankhya Labs, Invecas, Mindgrove Technologies, and Morphing Machines drive innovation in chip design. With DLI and C2S programmes, these startups receive financial incentives, mentorship, and access to EDA tools, enabling them to accelerate prototype-to-product transitions. Market Gap and Opportunity Analysis Despite major progress, a design-to-manufacturing gap persists in India's semiconductor landscape.

Opportunities:

- Government-backed programmes: C2S and DLI bridging R&D to market.
- Rising domestic demand: EVs, IoT, AI, 5G, and defence electronics.
- Global partnerships: Micron, Tower Semiconductor, Foxconn, and TSMC collaborations.
- Policy push: Atmanirbhar Bharat vision fostering self-reliance.

Opportunities and Challenges:

Opportunities:

- Expanding semiconductor demand across emerging sectors.
- Access to global EDA tools and fabrication support through ISM.
- Growing startup ecosystem and R&D base.

Challenges:

- High capital investment for fabs.
- Dependence on imported raw materials.
- Need for stronger integration between design, fab, and packaging.
- Global technology transfer limitations.

Conclusion:

The India Semiconductor Mission (ISM) and the Chips to Startup (C2S) program represent a turning point in India's technological journey. By combining policy support, industrial collaboration, and academic training, India is developing a complete semiconductor value chain — from ideation to production. The growing fabless design ecosystem, strengthened by DLI and C2S, ensures continuous innovation and skill development within the country.

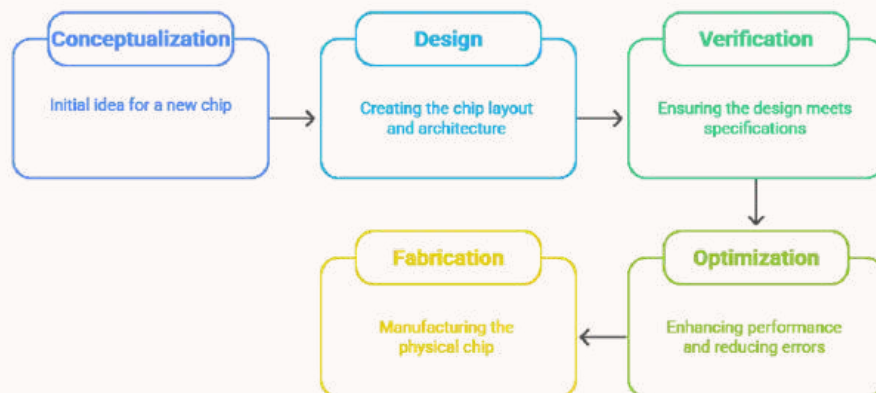


Dr. Shashi Kant Dargar
Associate Professor, ECE

EDA TOOLS AND WORK FLOW FOR MODERN SEMICONDUCTOR DESIGN

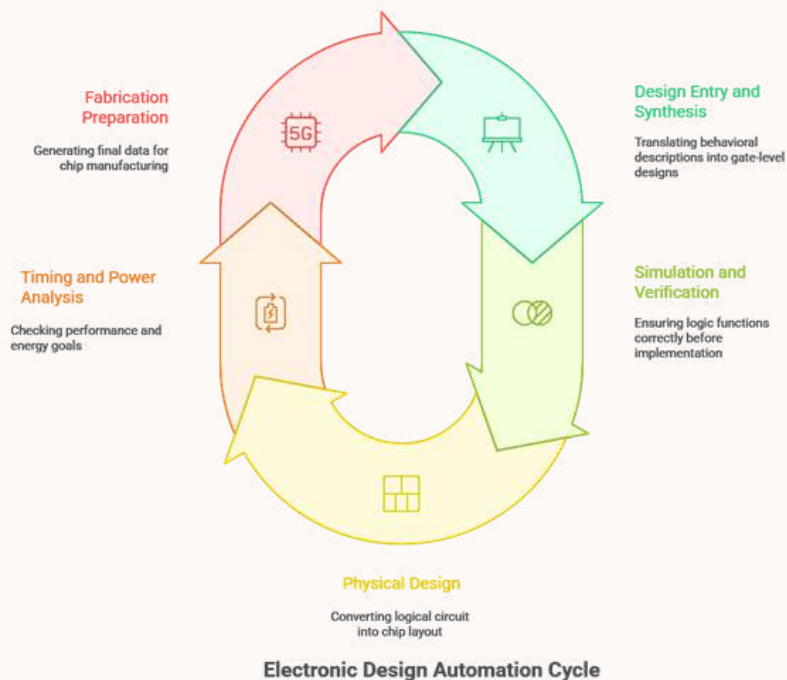
Introduction:

The modern world runs on silicon. From smartphones and laptops to automobiles and medical devices, semiconductors have become the backbone of innovation. Yet, behind every chip that powers these systems lies an intricate process of design, verification, and optimization — all made possible through Electronic Design Automation (EDA) tools. With shrinking transistor sizes and billions of components on board, manual designs are virtually impossible. The tools facilitate computational power, automation, and verification frameworks for taking a chip from concept to silicon with high performance, low power, and minimum errors.



Semiconductor Design Process with EDA Tools

EDA involves software that constitutes a suite of tools applied by engineers for the design, simulation, verification, and production of integrated circuits and systems-on-chip. In other words, EDA tools convert a designer's concept in high-level logic into a piece of physical layout that will eventually be produced on silicon wafers. EDA encompasses a wide array of functions Design Entry and Synthesis: Translation of behavioral descriptions-such as in VHDL or Verilog-into gate-level designs. Simulation and Verification: Ensuring the logic functions correctly before physical implementation. Physical Design: Converting the logical circuit into an actual chip layout (placement and routing). Timing and Power Analysis: Check if performance and energy goals are met. Fabrication Preparation: Creation of the final data used to fabricate the chip (GDSII or OASIS files).



Overview of Semiconductor Design Flow:

The semiconductor design flow is a methodical way to turn an idea into a working integrated circuit (IC). It starts with system-level design and specification, where engineers figure out how the chip will work and what it will look like. Then comes RTL (Register Transfer Level) design and verification, which makes sure that the logic is correct. The physical design stage uses synthesis, placement, and routing to turn the verified design into hardware. Finally, the design is checked for timing, verified, and taped out. This leads to the making and testing of the real chip. This flow makes sure that modern semiconductor devices work well, use little power, and are reliable.



Dr. Abha Dargar
Assistant Professor, ECE

Student Article

GAGANYAAN IS FUELING INDIA'S DIGITAL REVOLUTION

Abstract:

When India announced its first-ever human spaceflight mission, Gaganyaan, the world saw it as a bold leap into the elite club of spacefaring nations. But beneath the roar of rockets and the wave of national pride lies another story—one that is less visible but equally transformative. Gaganyaan is not just about sending astronauts into orbit; it is also quietly fueling a digital revolution at home.



The invisible backbone of Gaganyaan

At its heart, Gaganyaan is powered not only by engines and fuel tanks but also by algorithms, simulations, and data pipelines. Every stage of the mission—design, testing, and execution—relies on cutting-edge digital infrastructure:

- Simulations and Virtual Testing ensure spacecraft safety and performance without costly trial-and-error.
- Artificial Intelligence and Automation process mission data in real-time, securing astronauts against unforeseen risks.

- Big Data and Analytics manage terabytes of telemetry, enabling scientists to predict, adapt, and perfect the mission.

This makes Gaganyaan a digital-first mission, proving that India's space future rests as much on software and silicon as on rockets and fuel.

Astronaut Training:

If engineers rely on digital systems, astronauts live inside them. Gaganyaan's crew members undergo intensive digital training that mirrors the very skills needed in a 21st-century.

Virtual Reality Simulators recreate the spacecraft's cockpit, allowing astronauts to practice every possible emergency scenario.

AI-Driven Health Monitoring continuously tracks astronauts' vitals, teaching them to interpret and respond to digital readouts—skills that overlap with fields like telemedicine and wearable technology on Earth.



Robotics and Remote Operations give astronauts hands-on experience with semi-autonomous systems, reflecting the growing importance of human-AI collaboration in industry.

By preparing astronauts to think critically, respond quickly, and adapt seamlessly to new digital environments, Gaganyaan underscores how digital literacy is not optional—it is survival.

Inspiring a Digital Nation:

The impact of Gaganyaan extends beyond ISRO's laboratories, reaching classrooms, homes, and villages across India. Much like the Apollo program inspired a generation in the United States, Gaganyaan is sparking curiosity among India's youth:

Accessible Space Online: ISRO shares mission updates, visuals, and learning resources freely, bringing space into the public's hands.

Digital Classrooms: Teachers use Gaganyaan to inspire projects in coding, robotics, and simulation-based learning.

bridging the Divide: For rural and underserved communities, the mission proves that advanced digital tools are not out of reach but pathways to opportunity.

In this way, Gaganyaan is more than a rocket—it is a national classroom that makes digital learning real and aspirational.

Digital Literacy as self-reliance

India's vision of Aatmanirbhar Bharat (self-reliant India) resonates deeply with both Gaganyaan and digital literacy. Just as building our own spacecraft frees the nation from dependence on foreign launch services, building a digitally skilled population frees us from reliance on imported knowledge economies.

Every child who learns coding, every teacher who integrates digital tools, and every citizen who becomes digitally literate contributes to this self-reliance.

How Gaganyaan is Fueling India's Digital Revolution

When India announced its first-ever human spaceflight mission, Gaganyaan, the world saw it as a bold leap into the elite club of spacefaring nations. But beneath the roar of rockets and the wave of national pride lies another story—one that is less visible but equally transformative. Gaganyaan is not just about sending astronauts into orbit; it is also quietly fueling a digital revolution at home, bridging the gap between advanced space technology and everyday life on Earth.

The Gaganyaan rocket represents India's bold step into human spaceflight, but the mission is also deeply intertwined with the digital revolution. Depicting the launch with AI icons, binary streams, VR headsets, and data graphs blending into the sky symbolizes the fusion of space exploration and digital technology



Conclusion

Gaganyaan is more than a technological milestone—it is a symbol of India's digital awakening. By blending space exploration with digital innovation, the mission showcases how science and technology can uplift an entire nation. As astronauts prepare to carry the tricolor into space, they also carry a powerful message back home: that mastering digital literacy is as vital to India's future as mastering space itself. Just as Gaganyaan is a leap into the cosmos, digital literacy is India's leap into self-reliance. Together, they chart a path toward a future where every citizen can take part in the country's scientific, technological, and digital journey.

Sai Siddarth
9923005219

EVOLUTION OF 6G COMMUNICATION (REDEFINING CONNECTIVITY WITH INTELLIGENCE, SPEED, AND UBIQUITY)

Abstract

6G will move beyond 5G with ultra-fast speeds, near-zero latency, and universal connectivity, driven by terahertz, AI-native networks, reconfigurable surfaces, and quantum technologies. It promises transformative applications like XR, holography, and the tactile internet, while global research addresses key architectural and integration challenges..

Introduction:

The history of wireless communication is marked by generational leaps, each transforming how societies connect and share information. From the analog voice systems of 1G to the gigabit broadband of 5G, each generation has built upon its predecessor, addressing new demands and technological limitations.

1G [1980s] : Analog voice transmission with limited mobility and security.

2G [1990s] : Digital voice (GSM, CDMA) with SMS and basic data services.

3G [2000s] : Enhanced data speeds enabling mobile internet and multimedia.

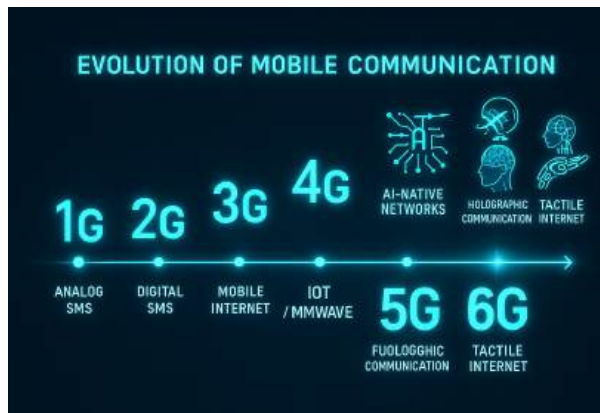
4G [2010s] : High-speed data (LTE) supporting HD video, VoIP, and apps.

5G [2020s] : Enhanced mobile broadband (eMBB), ultra-reliable low-latency communications (URLLC), and massive machine-type communications (mMTC).

Technical Overview of 6G:

- **Data Rates:** Up to 1 Tbps
- **Latency:** Less than 0.1 milliseconds
- **Device Density:** Over 10 million devices per square kilometer
- **Reliability:** >99.99999%
- **Mobility Support:** Up to 1000 km/h

6G builds on the evolution from 1G to 5G, delivering terabit data rates, ultra-low latency, massive connectivity, and extreme reliability.



Timeline of Mobile Communication Generations from 1G to 6G

Key Enabling Technologies

The vision of 6G is driven by a set of revolutionary enabling technologies that push the boundaries of speed, intelligence, and security in communication systems. From harnessing the power of terahertz waves for ultra-broadband data transfer to deploying intelligent surfaces that reshape signal propagation, these innovations form the backbone of future networks.

1. **Terahertz Communication (THz):** Operating in the 100 GHz to 10 THz band, THz waves enable ultra-broadband channels ideal for Tbps data rates. Challenges include high path loss, absorption, and hardware complexity.
2. **Reconfigurable Intelligent Surfaces (RIS):** RIS are programmable meta surfaces that dynamically control electromagnetic wave propagation, enhancing signal quality, coverage, and spectral efficiency.
3. **AI/ML-Native Networking:** 6G is envisioned as AI-native, with integrated machine learning for real-time resource allocation, fault prediction, and self-optimizing networks.
4. **Communication and Security:** Quantum key distribution (QKD) and entanglement-based communication offer unbreakable encryption and new modes of secure information transfer.



Key Technologies, Benchmarks, Challenges, and Applications of 6G Communication

Challenges in 6G Deployment

This include spectrum availability, as securing and regulating THz bands for commercial use presents regulatory and technical hurdles. Infrastructure scalability is another concern, since the deployment of ultra-dense networks with RIS, edge computing, and THz base stations requires significant capital and planning.

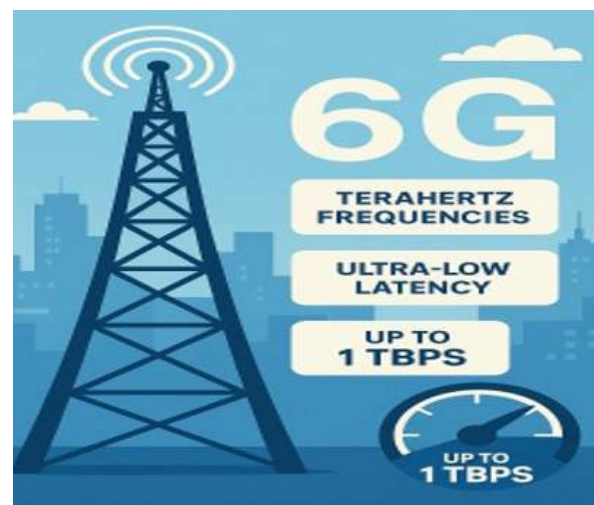
Energy efficiency also poses a challenge, because higher frequencies and AI workloads increase power consumption, making green communication technologies essential. Finally, hardware maturity remains an obstacle, as reliable and cost-effective THz components, RIS panels, and quantum devices are still in development.

Potential Applications of 6G

- **Tactile Internet:** Supports ultra-responsive feedback for remote control in industrial automation and robotic systems.
- **Real-Time Remote Surgery:** Combines ultra-low latency with AI and XR for precise, high-risk medical procedures conducted remotely.
- **Metaverse and Immersive XR:** 6G will enable high-fidelity, real-time interaction in virtual worlds with full sensory immersion.
- **Autonomous Systems:** Vehicles, drones, and industrial robots benefit from edge intelligence and robust connectivity.

Conclusion

The evolution from 1G to 6G represents more than just an increase in speed or capacity—it is a fundamental reimagining of how communication networks interact with the physical and digital worlds. 6G is poised to usher in an era of hyper-connectivity, where intelligent systems operate seamlessly through ultra-reliable, low-latency, and secure channels powered by terahertz frequencies, AI-native orchestration, and quantum resilience.



6G Communication Performance Highlights

Unlike its predecessors, 6G will not merely connect people—it will connect intelligence. Networks will become predictive, autonomous, and deeply embedded in every facet of society, enabling applications once thought to be the realm of science fiction: real-time holographic interaction, human-robot tactile interfaces, and fully immersive digital realities. These advances will redefine the boundaries of innovation in healthcare, education, manufacturing, entertainment, and governance.

As we stand on the brink of this next technological epoch. Continued research, ethical foresight, and sustainable development will be critical in ensuring that 6G fulfils its transformative promise for generations to come.

B.Neelakanteswar Reddy
9923005150

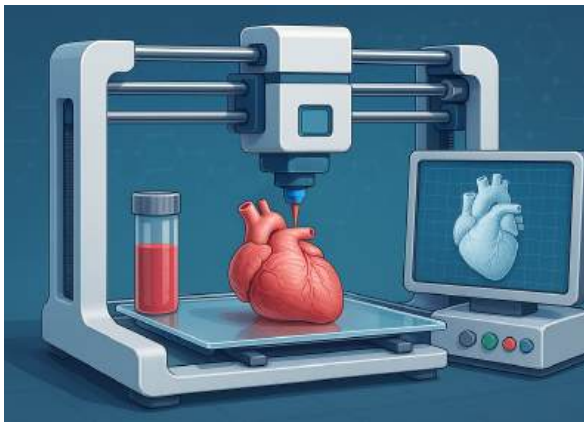
3D BIOPRINTING MANUFACTURING HUMAN TISSUES AND ORGANS

Abstract

3D bioprinting uses bio-inks made of living cells and biomaterials to create human tissues and organs layer by layer. It aims to address organ shortages and advance personalized medicine by precisely controlling cell organization. Though challenges like vascularization, scalability, and biocompatibility remain, ongoing research shows promise for regenerative medicine, drug testing, and transplantation.

What is 3D Bioprinting?

3D bioprinting is an advanced manufacturing process that involves the creation of three-dimensional, biologically functional structures using computer-aided design (CAD) models. This technology allows for the precise placement of cells, biomaterials, and growth factors to form tissue-like structures that closely mimic the natural architecture and functionality of human tissues and organs. The core idea is to "print" living materials layer by layer to build complex biological systems.



Advanced 3D bioprinter in action layer-by-layer construction of a bioengineered human heart.

Three key stages in Bio-Printing:

Pre-Bioprinting: This initial stage involves preparation and planning. Medical imaging techniques such as MRI or CT scans are used to design a detailed blueprint of the tissue or organ to be printed. Based on this model, suitable bio-inks composed of living cells and supportive biomaterials are selected and prepared. The choice of bio-ink depends on the type of tissue being printed, its mechanical properties, and biological compatibility.

Bioprinting: In this stage, the actual printing process occurs. Specialized bioprinters deposit the bio-ink in a precise, layer-by-layer manner using techniques such as extrusion, inkjet, or laser-assisted printing. The deposition follows the digital blueprint to replicate the complex internal structure of natural tissues. The goal is to ensure the correct spatial arrangement of different cell types and materials for optimal function.

1.

Post-Bioprinting: After printing, the newly formed structures are not yet fully functional. They require a maturation period where they are placed in bioreactors—controlled environments that supply the necessary nutrients, oxygen, and mechanical stimuli. This stage allows the cells to grow, interact, and develop into fully functional tissues with properties similar to their natural counterparts.

Breakthroughs and Challenges:

Bioprinting has advanced in creating simpler tissues like skin, cartilage, and bone, benefiting regenerative medicine. However, printing complex organs such as the heart, liver, and kidneys remains challenging due to the need for intricate vascular networks. Researchers are exploring vascularization techniques and microfluidic systems to mimic blood flow, but overcoming these hurdles is crucial before clinical use of bioprinted organs.

Applications in Healthcare :

3D bioprinting is transforming the landscape of healthcare by offering innovative solutions for tissue regeneration, transplantation, and pharmaceutical research. Its ability to fabricate patient-specific biological structures opens up numerous life-saving applications, some of which are outlined below

Skin Grafts: One of the most immediate applications of 3D bioprinting is in treating burn victims and patients with severe skin injuries. By using a patient's own skin cells to print customized grafts, the risk of immune rejection is significantly reduced. These bio-printed skin grafts can be tailored in size, shape, and cellular composition to match the wound precisely, accelerating healing and improving cosmetic outcomes.

Cartilage and Bone Regeneration: Bioprinted cartilage and bone tissues offer promising alternatives to traditional orthopedic and dental implants. These printed structures can be engineered to match the mechanical properties of native tissues, allowing for better integration with the body. Applications include joint repair, facial reconstruction, and spinal fusion procedures, where customized implants can improve patient outcomes and reduce recovery time.

Organ Patches: Instead of printing entire organs, researchers are currently focusing on creating smaller, functional tissue patches for organs such as the heart, liver, or kidneys. These patches can be used to repair damaged tissue, restore partial function, or support the healing process. For example, cardiac patches with living cells can be applied to the heart muscle after a heart attack to improve regeneration and function.

Drug Testing and Development: 3D bio-printed tissues offer a more ethical and accurate alternative to animal testing. Scientists can create tissue models that closely mimic human physiology, allowing them to study how drugs interact with specific cell types or tissues. This improves the accuracy of drug toxicity and efficacy testing, speeds up the drug development process, and reduces reliance on animal models, ultimately leading to safer and more effective treatments.

Future Prospects

As research in biomaterials, stem cell technology, and computational modeling progresses, the future of 3D bioprinting looks increasingly promising. Scientists envision a time when the global shortage of donor organs could be eliminated through the ability to bioprint fully functional, patient-specific organs. These organs would be created using the patient's own cells, minimizing the risk of rejection and eliminating the need for lifelong immunosuppressive drugs. Beyond transplants, bioprinting could also enable the development of disease models for personalized drug testing, further advancing precision medicine. With continued innovation and interdisciplinary collaboration, 3D bioprinting has the potential to redefine the future of healthcare.



A digital blueprint guiding the 3D bioprinter in fabricating a human heart using living cells.

Conclusion:

3D bioprinting, merging electronics, biology, and material science, offers groundbreaking solutions to organ shortages and personalized medicine. While challenges like vascularization and functional integration remain, rapid progress indicates large-scale production of tissues and organs is achievable. For Electronics and Communication Engineering (ECE) students, this field showcases how electronics contribute through precise control systems, embedded sensors, and monitoring tools, making ECE a vital part of future biomedical innovations.

Aakash Bala Prajapathi S
9923005183

ROLE OF VLSI IN THE EVOLUTION OF AI

(WHERE SILICON MEETS INTELLIGENCE: VLSI AT THE HEART OF AI PROGRESS)

Abstract

Very Large Scale Integration (VLSI) has been pivotal in transforming Artificial Intelligence (AI) from a theoretical field into a ubiquitous, real-time computing paradigm. As AI workloads have evolved from algorithmic novelty to mission-critical applications, the demand for specialized silicon has surged. This paper explores the deep synergy between VLSI and AI, examining how innovations in semiconductor design, including low-power architectures, neural network accelerators, and in-memory computing, have enabled scalable and efficient AI systems. Through case studies such as Google's TPU, Intel's Loihi, and Apple's Neural Engine, we highlight real-world implementations. The article also discusses major design challenges and outlines emerging trends such as neuromorphic computing and chiplet-based architectures that will shape the future of AI hardware.

Understanding VLSI and Its Relevance in Modern Computing

Very Large Scale Integration (VLSI) refers to the process of integrating millions (and now billions) of transistors on a single silicon chip. Introduced during the late 1970s, VLSI enabled the development of microprocessors, memory units, and complex SoCs (System on Chips) that power everything from smartphones to supercomputers.

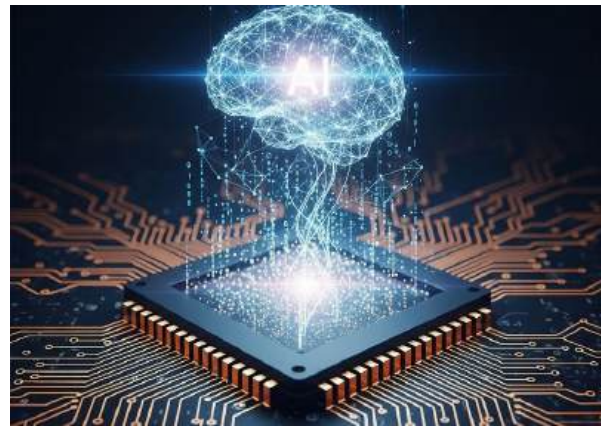
In the context of modern computing, VLSI is not just about packing more transistors; it's about intelligent design—optimizing power, performance, area, and cost (PPAC) for diverse and demanding workloads. With the rise of AI, the design philosophy of VLSI has evolved to support domain-specific accelerators that can process massive parallel computations efficiently.

Evolution of AI Workloads and the Need for Specialized Silicon:

- AI workloads, particularly those involving deep learning, exhibit high data parallelism and computational intensity. Traditional CPUs struggle with these characteristics due to their general-purpose nature and limited parallelism.

Key Trends Driving Silicon Specialization:

- **High throughput and low latency:** AI models (e.g., CNNs, RNNs, Transformers) require billions of multiply-accumulate (MAC) operations per inference.
- **Memory bottlenecks:** Frequent memory access in large models creates a significant energy and bandwidth challenge.
- **Edge deployment:** AI at the edge (e.g., in autonomous vehicles, wearables, drones) demands compact, low-power hardware.



Conceptual visualization of artificial intelligence emerging from advanced silicon Technology

Silicon Solutions:

GPUs (Graphics Processing Units) - Efficient for training and inference due to massive parallelism.

FPGAs (Field Programmable Gate Arrays): Configurable accelerators for customized AI workflows.

ASICs (Application Specific Integrated Circuits): Tailored for specific AI operations, offering high efficiency and low power.

VLSI Contributions to AI Chip Design:

Low-Power Architectures - VLSI enables the design of energy-efficient circuits by leveraging advanced techniques such as clock gating, dynamic voltage-frequency scaling (DVFS), and near-threshold computing.

THE EVOLUTION AND FUTURE OF CYBER CRIMES

(A LOOK INTO THE DARK SIDE OF THE DIGITAL AGE)

Abstract

Cybercrime has rapidly evolved from basic acts of curiosity to sophisticated, organized operations that threaten individuals, corporations, and governments worldwide. This article examines the historical progression of cybercrimes, beginning from early hacking incidents to modern threats such as ransomware, phishing, and state-sponsored cyber espionage. It also explores the impact of emerging technologies like Artificial Intelligence (AI), the Internet of Things (IoT), and quantum computing on the cybersecurity landscape. As digital integration deepens, the article emphasizes the urgent need for global cooperation, technological innovation, and increased public awareness to combat future cyber threats effectively.

The Early Days of Cybercrime Key Points

Cybercrime in the 1970s–1980s:

Initially driven by curiosity and experimentation rather than financial or political motives.

1971 – Phone Phreaking Incident:

A group discovered that a toy whistle from Cap'n Crunch cereal emitted a 2600 Hz tone, used by AT&T systems.

1986 – Brain Virus:

Created by two brothers in Pakistan.
Spread via infected floppy disks, targeting the boot sector, included contact information.

1990s – Rise of the Internet:

Internet introduced remote and anonymous access to systems.
Enabled cybercriminals to attack targets globally without physical contact.

Early Online Exploits:

Web page defacements, spam emails, and unauthorized intrusions became common.
Motivations were often about peer recognition rather than monetary gain.

Shift in Cybercrime Landscape:

Laid the foundation for the professional, organized, and profit-driven cybercrime seen in later decades.
Highlighted the potential of digital tools to cause widespread disruption.



A timeline of digital threats – from early hackers to AI-driven cybercrime in the coming decades

The Rise of Organized Cybercrime

Growth of Financially Motivated Attacks

Phishing scams tricked users into revealing passwords, banking details, and personal information. As a result, identity theft became widespread, allowing criminals to open bank accounts, apply for credit, and commit fraud. At the same time, online financial frauds increased, targeting both individuals and corporations.

Advanced Cybercrime Tools

Trojans, disguised as legitimate programs, were used to steal data or provide remote access. Ransomware attacks encrypted users' data and demanded payment for release, while botnets networks of infected devices launched large-scale attacks like Distributed Denial of Service (DDoS).

Impact of Commercialization

The entry barriers for cybercrime participation were lowered, making it easier for more people to get involved. This led to an increased scale of attacks, where millions of systems could be compromised from anywhere in the world. As a result, cybercrime evolved into a multi-billion-dollar underground industry.

Global Impact :

Cybercrime inflicts trillions of dollars in losses on the global economy each year, with the costs steadily rising as attacks become more frequent and sophisticated. These losses include direct financial theft, ransom payments, data recovery expenses, legal penalties, business downtime, and long-term reputational damage to companies and governments.

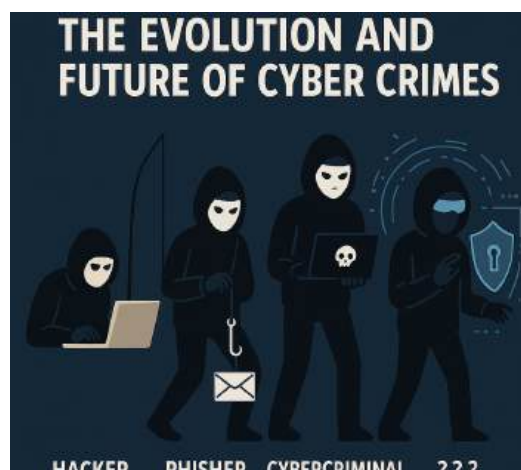
Beyond financial damage, cyberattacks have the potential to severely disrupt critical services. In the healthcare sector, ransomware attacks have forced hospitals to cancel surgeries, delay treatments, and revert to outdated manual record-keeping systems, putting lives at risk. Attacks targeting infrastructure such as power grids, transportation systems, and communication networks can bring entire regions to a standstill and endanger public safety.

The Future of Cybercrime :

As technology continues to advance at an unprecedented pace, the nature of cybercrime is expected to become even more complex and dangerous. One of the most significant upcoming threats is the emergence of quantum computing. Unlike classical computers, quantum machines will be capable of processing massive amounts of data simultaneously. This means that they could potentially break current encryption standards, rendering most of today's cybersecurity protocols obsolete. As a result, organizations and governments will need to invest in post-quantum cryptography to protect sensitive data from future quantum-based attacks. Another growing concern is the misuse of deepfake technology. With the help of artificial intelligence, cybercriminals can now create highly realistic fake videos and audio clips that mimic real people, including public figures, CEOs, and politicians. These manipulated media can be used to spread misinformation, commit fraud, manipulate public opinion, or even authorize illegal transactions, making it increasingly difficult to determine what is real and what is not.

Countermeasures and the Way Forward:

In response to the growing threat of cybercrime, governments, corporations, and cybersecurity professionals around the world are making significant investments in defense strategies to protect digital infrastructure and sensitive data. As cyber threats become more advanced, so must the technologies and policies used to combat them. One of the primary countermeasures is the implementation of stronger encryption standards.



From simple hacks to advanced digital intrusions uncover the transformation of cybercrime through the decades and what lies ahead

Conclusion:

Cybercrime is no longer just a technical issue—it is a social, economic, and political challenge. As we move further into the digital age, staying ahead of cyber threats will require not only advanced technologies but also global collaboration and responsible digital behavior. The future of cybersecurity lies in proactive defense, ethical innovation, and continuous education.

Shafra Mariyam I
9923005221

ROLE OF AUGMENTED REALITY IN EDUCATION

Abstract

Augmented Reality (AR) is revolutionizing the education landscape by integrating digital enhancements into real-world environments, creating interactive and immersive learning experiences. This technology bridges the gap between theoretical knowledge and practical understanding by enabling students to engage with 3D models, simulations, and real-time data. AR supports personalized learning, enhances student motivation, and improves retention of complex concepts. As AR continues to evolve, it holds immense potential to reshape traditional teaching methods and create more dynamic, student centered educational environments.

What is Augmented Reality?

Augmented Reality (AR) overlays digital images, information, or animations onto the real world through devices like smartphones, tablets, or AR headsets. Unlike Virtual Reality (VR), which creates a fully digital environment, AR blends digital content with reality in real time. In education, it helps bring textbooks to life, visualize abstract ideas, and create interactive learning experiences without taking learners out of their surroundings.

Applications of AR in Education :

Science and Biology:

AR enables students to explore the human body in 3D, observe organs from different angles, conduct virtual dissections without harming any life, and visualize molecular or chemical reactions in real-time. This fosters a deeper understanding of otherwise abstract scientific phenomena.

Engineering and Technology:

AR allows engineering students to interact with digital models of engines, electronic circuits, and mechanical systems, offering a hands-on feel without needing physical components. Complex simulations can be overlaid on real surfaces, aiding in design thinking and troubleshooting.



Exploring global knowledge through Augmented Reality
bringing textbooks to life in the classroom

History and Geography:

Through AR, students can walk through ancient ruins, witness historical events unfold, or interact with 3D maps and globe models. This immersive experience enhances historical empathy and geographic understanding, making learning more memorable.

Medical Training:

Medical students benefit from realistic simulations of surgeries, anatomy, and emergency scenarios. AR can provide layered views of body systems, helping them grasp the spatial relationships of organs and practice procedures in a risk-free environment.

Interactive Textbooks:

Traditional textbooks become dynamic when integrated with AR. When scanned with a device, they can display animated diagrams, audio explanations, or interactive quizzes, encouraging self-paced and engaging learning.

Language Learning:

AR flashcards and apps can provide pronunciation, translations, and context for words, while real-world object recognition enhances vocabulary acquisition through visual association.

Benefits of AR in Education :

Enhances engagement and interest:

AR transforms traditional lessons into immersive experiences, capturing students' attention and making learning more enjoyable and interactive.

Promotes experiential learning and better retention:

By allowing students to actively interact with content, AR fosters hands-on learning, which leads to improved memory retention and deeper understanding.

Makes abstract and complex concepts easier to understand:

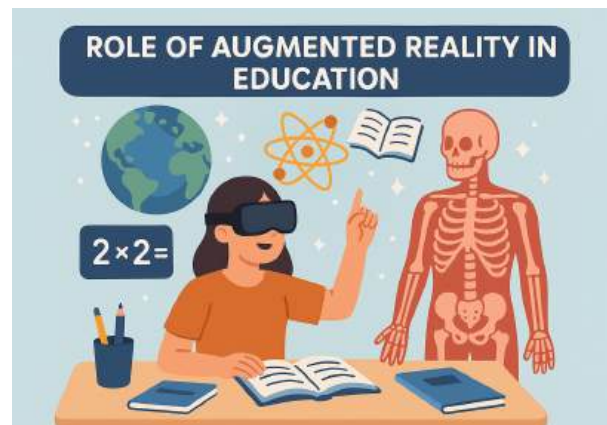
Difficult subjects like anatomy, physics, or mathematics become more accessible when learners can visualize and manipulate 3D models or animations.

Provides a safe and controlled environment for practice:

Students can simulate risky or expensive experiments like surgeries or lab procedures without any real-world consequences, encouraging trial and error without fear.

Challenges and Considerations :

Despite its many advantages, augmented reality (AR) in education faces several challenges. High costs of AR-enabled devices can burden institutions with limited budgets, while many educators lack the technical training needed to use these tools effectively. The limited availability of curriculum-aligned AR content also makes it difficult to find relevant, engaging materials for students. Concerns about reliability and the need for ongoing technical support add to the obstacles. However, as AR becomes more affordable and accessible, investments in teacher training and tailored content are gradually paving the way for wider adoption in classrooms. Collaboration among educators, developers, and policymakers is essential to address these challenges, and continued innovation will expand learning opportunities across different subjects.



Immersive AR learning: Visualizing science, math, and anatomy for deeper understanding and engagement.

Conclusion:

Augmented Reality is transforming education by bridging the gap between theory and practice. For electronics and communication engineering students, AR offers new ways to learn and explore concepts through interactive simulations. It also creates opportunities for research in areas like signal processing, sensor design, and user interfaces. As education becomes more interactive, AR is leading the way toward a more engaging and effective learning experience. With its growing accessibility, AR is set to become an integral part of modern classrooms. Its continued development promises to shape the future of education and research in powerful ways. AR can also encourage collaborative learning by enabling students to work together on complex problems in a shared virtual environment. By blending theory, practice, and innovation, it empowers learners to develop skills that are directly applicable to real-world challenges.

Aakash Bala Prajapathi S
9923005183

Future of Digital Currency and Its Challenges

Introduction

Digital currency is revolutionizing the global financial system. The concept, which began as an experimental technology with Bitcoin in 2009, has evolved into a serious contender for replacing physical money. The adoption of digital currencies by both individuals and institutions is accelerating, driven by the promise of greater efficiency, security, and accessibility. Central banks, too, are entering the race by developing their own digital currencies, commonly referred to as Central Bank Digital Currencies (CBDCs). This article explores the various dimensions of digital currencies their types, benefits, challenges, and the future they are shaping.

Types of Digital Currency

Digital currency can be broadly classified into three categories:

Cryptocurrencies :

These are decentralized currencies like Bitcoin, Ethereum, and Litecoin that operate on blockchain technology without any central authority.

Stablecoins :

These are digital currencies backed by real-world assets like the US dollar or gold. Examples include USDT (Tether) and USDC.

CBDCs :

Issued and regulated by governments or central banks, CBDCs are the digital form of a country's official currency (e.g., digital rupee, digital yuan).

Technological Foundation

Blockchain and distributed ledger technology (DLT) form the backbone of most digital currencies. These systems ensure transparency, immutability, and security. Smart contracts self-executing code on the blockchain further enhance functionality by automating financial agreements. In addition, technologies such as zero-knowledge proofs and quantum cryptography are being explored to increase security and scalability.

Economic Impact

The emergence of digital currencies, particularly Central Bank Digital Currencies (CBDCs), has the potential to significantly reshape global economic models and monetary policy. Governments could use CBDCs to execute real-time taxation and direct stimulus payments, enhancing transparency, efficiency, and fiscal control. At the same time, traditional banks may face disruption as people shift toward decentralized financial services (DeFi), reducing reliance on intermediaries.



Social and Cultural Shifts:

Digital currencies are transforming society by improving financial inclusion, especially in underserved regions where traditional banking is limited. With mobile wallets and internet access, people can store money, send payments, and access financial services without visiting a bank. This empowers rural communities and promotes economic participation. However, challenges such as the digital divide, lack of financial literacy, and concerns over data privacy and surveillance pose risks. To fully realize the benefits, it is essential to educate users and ensure secure, accessible digital infrastructure.

Regulatory Landscape:

The regulatory landscape surrounding digital currencies is complex and rapidly evolving, with governments around the world adopting vastly different approaches. Some countries, like El Salvador, have taken bold steps by recognizing Bitcoin as legal tender, aiming to promote financial innovation and attract crypto investment. Others, including China and Algeria, have imposed strict bans on cryptocurrency trading and mining, citing concerns over financial stability, capital flight, and illegal activities. Many nations, however, fall somewhere in between working to create regulatory frameworks that balance innovation with consumer protection and financial security. The absence of global regulatory consensus has created a fragmented environment that complicates cross-border transactions, compliance, and enforcement..

Environmental Impact

The environmental impact of cryptocurrencies, especially those using Proof of Work (PoW) like Bitcoin, has raised global concerns due to their high energy consumption.

Mining operations require powerful hardware and large amounts of electricity, sometimes comparable to the energy use of entire nations. Critics argue that this environmental toll outweighs the benefits, especially as adoption grows. In response, the industry is shifting toward more sustainable alternatives such as Proof of Stake (PoS), which drastically reduces energy usage by eliminating mining. Ethereum's move to PoS, for example, cut its energy consumption by over 99%. Additionally, efforts are being made to power mining with renewable energy sources and to implement carbon offset programs that reduce the ecological footprint. As awareness increases and regulations tighten, the digital currency sector is gradually evolving toward greener, more sustainable practices



“Digital currency expands with growth and innovation, but faces hurdles in security, regulation, and market volatility.”

Conclusion

The future of digital currency is both promising and complex. While the technology offers numerous benefits in terms of speed, cost, and accessibility, it also brings unprecedented challenges related to regulation, security, and ethics

Sai Siddarth
9923005219

Featured Article: A Glimpse into Brain-Inspired Chips

Abstract:

Neuromorphic computing represents a paradigm shift in the design of intelligent systems, inspired by the architecture and functionality of the human brain. Unlike conventional von Neumann architectures, neuromorphic systems integrate memory and processing units to enable massively parallel, low-power, and adaptive computation. By leveraging spiking neural networks (SNNs), event-driven architectures, and emerging hardware technologies such as memristors and phase-change materials, neuromorphic platforms promise breakthroughs in real-time learning, pattern recognition, and edge computing.

The Neuromorphic computing:

Neuromorphic computing is designed to emulate the organizational and functional principles of the human brain, encompassing both hardware and algorithms. Unlike von Neumann systems that separate memory and processing, neuromorphic designs integrate both, often using spiking neural networks (SNNs)—where information is passed via discrete “spikes,” much like neuronal firing.

Key players & Innovations:

Hardware implementations:

- Intel Loihi, IBM TrueNorth, and BrainChip Akida offer neuromorphic processors optimized for spiking computation.
- University of the People IBM SpiNNaker, a massively parallel system from the Human Brain Project, simulates spiking neural networks at large scales.

- Emerging materials like memristors, ferroelectrics, and phase-change memory enable co-located memory and processing capabilities.
- If engineers rely on digital systems, astronauts live inside them. Gaganyaan’s crew members undergo intensive digital training that mirrors the very skills needed in a 21st-



Core Advantages

Energy Efficiency: Uses significantly less power than traditional AI models—ideal for edge devices like wearables and sensors.

- **Real-Time Adaptability:** Supports on-the-fly processing and learning—critical for robotics, autonomous systems, and smart sensors.
- **Parallel Processing & Robustness:** Handles noisy or incomplete data effectively via distributed, spike-driven computation. Baeldung on Kotlin InformationWeek Educba

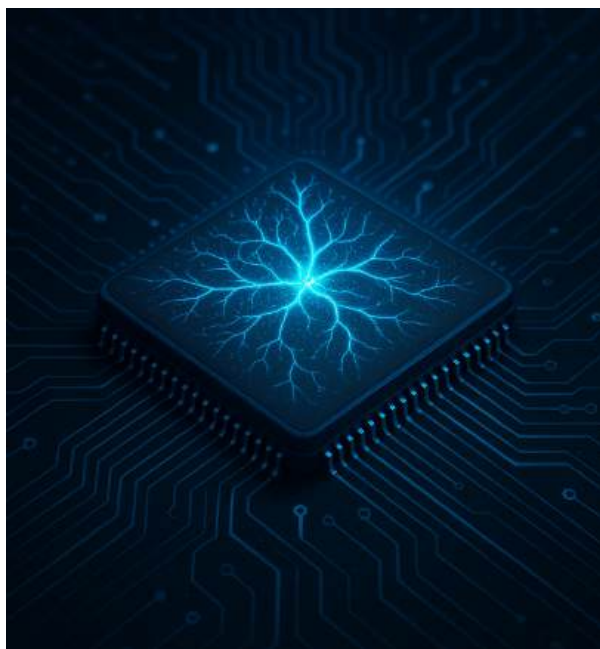
Use Cases & Applications:

IoT & Edge AI: Enables always-on, low-power devices that make autonomous decisions.

Robotics & Autonomous Systems: Enhances navigation, sensor fusion, and real-time decision-making.

Healthcare & Prosthetics: Powers brain-computer interfaces, prosthetic control, and real-time diagnostics.

University of the People
Detection: Identifies threats through pattern recognition and spiking behavior



By leveraging spiking neural networks (SNNs), event-driven architectures, and emerging hardware technologies such as memristors and phase-change materials, neuromorphic platforms promise breakthroughs in real-time learning, pattern recognition, and edge computing.

Challenges Ahead

Tooling & Programming Complexity: Developing SNNs requires new skills and frameworks—still maturing.

Lack of Standards: Fragmented hardware/software ecosystems hinder interoperability.

- Limited Commercial Adoption: While trending toward mainstream, applications remain niche
- Hardware Development Hurdles: Scaling reliable neuromorphic hardware remains complex. Educba arXiv

Conclusion:

Neuromorphic computing is pushing the boundaries of how machines learn and interact with the world—bringing us closer to brain-inspired efficiency, adaptability, and intelligence. While hurdles remain, developments like the Innatera Pulsar microcontroller, on-chip learning techniques, and innovative sensors mark a turning point.

Shaik Jafiya Naz
9923005039

INSPIRING PERSONALTIES



Malhar Kalambe, a young environmentalist from Mumbai, is the founder of the youth-led movement "Beach Please." Starting with a mission to clean Dadar beach, his organization has cleared millions of kilograms of waste from Mumbai's coastlines and rivers. Kalambe's relentless dedication has mobilized thousands of volunteers and earned him UN recognition, showcasing the power of consistent, citizen-led action in tackling urban pollution.

Divya Deshmukh, a chess prodigy from Nagpur, is one of India's brightest stars in the sport. She achieved the titles of Woman Grandmaster (WGM) and International Master (IM) at a young age, securing multiple national championships. A key member of the 2022 Chess Olympiad bronze-winning team, her tactical acumen and composure make her a leading figure in the new generation of Indian chess champions.



Sheetal Devi, a para-archer from Jammu and Kashmir, has become a global icon of resilience. Born without arms (phocomelia), she is the world's first female archer to compete internationally using her feet. Her extraordinary performance at the 2022 Asian Para Games, where she won two gold medals, has shattered perceptions of disability and established her as a powerful symbol of human potential and determination.

Dr. Sivaranjani Santosh, is a practicing obstetrician and gynaecologist who leverages social media to champion women's health education. She uses her platform to debunk medical myths and provide clear, evidence-based information on taboo subjects like menstruation and gynaecological health. Her accessible approach empowers millions of women to understand their bodies and advocate for their well-being, fighting misinformation in the digital age.







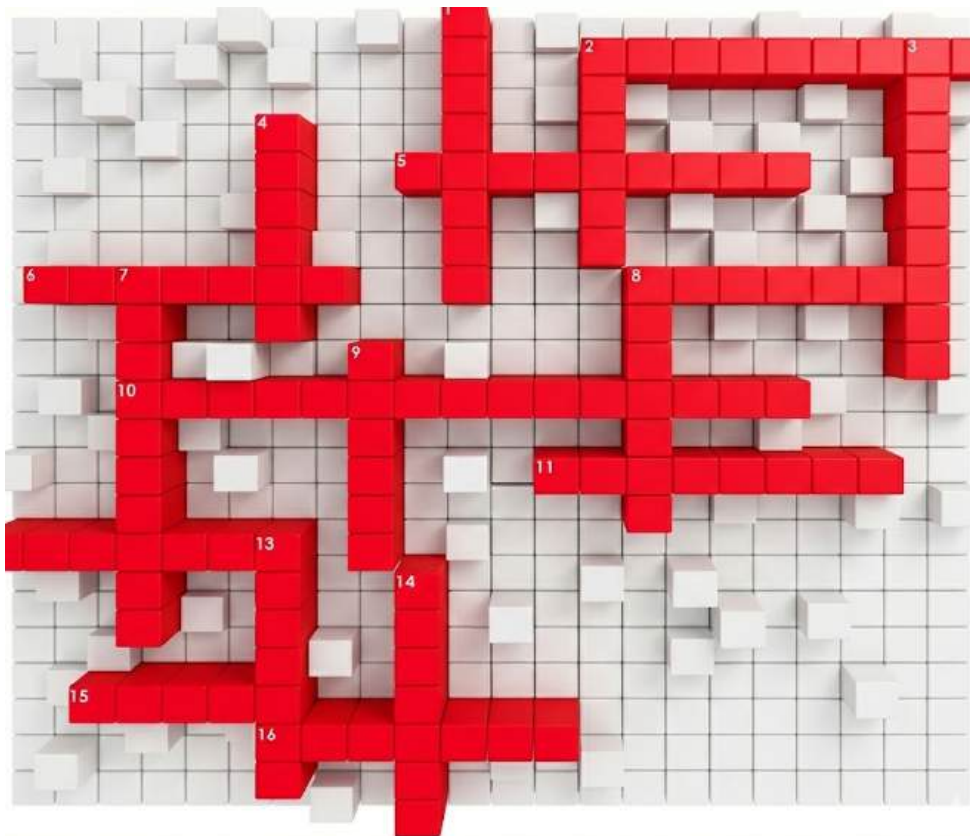
CROSSWORD PUZZLES

Across

2. A diagram that shows the electrical connections of the electronic components
5. Current is considered to be the movement of _____.
6. A voltage source that converts chemical energy to electrical energy
8. A flow of electric charge
10. A characteristic of a secondary cell
11. A material that is composed of a mixture of elements
12. The term used to designate electrical pressure
15. A short circuit will have a _____ current flow.
16. The part of an atom that has no electric charge

Down

1. A voltmeter is used in _____ with the circuit.
2. A device that opens or completes an electrical path
3. A material that opposes the movement of free electrons
4. One coulomb passing a point in one second
7. A resistive component that is designed to be temperature sensitive
8. A unit of charge that contains 6.25×10^{18} electrons
9. An atom's atomic number is determined by its number of _____.
13. A substance that is found only in its pure form
14. It is used to measure current.





FIND THE WORD?

E	Y	B	D	C	M	R	R	A	T	T	E	N	U	A	T	I	O	N	V
N	T	J	L	R	I	A	P	L	A	I	T	N	E	R	E	F	F	I	D
I	I	T	O	W	C	F	E	T	A	R	T	S	I	U	Q	Y	N	X	I
L	C	C	H	A	R	A	C	T	E	R	I	S	T	I	C	O	C	G	N
E	O	A	E	V	O	N	I	N	V	H	T	R	S	T	C	O	T	R	L
P	L	X	L	E	S	Q	O	S	P	A	O	M	R	U	N	R	A	O	D
I	E	N	P	G	T	B	W	I	T	O	I	E	R	T	A	S	G	U	Z
P	V	I	M	U	R	R	D	E	T	T	T	R	R	N	T	A	S	P	T
T	E	G	A	I	I	N	S	L	H	U	E	O	S	U	I	M	T	V	I
R	S	R	S	D	P	P	O	C	R	N	L	F	B	N	S	P	R	E	W
A	A	A	O	E	A	C	H	N	T	L	E	O	M	L	O	L	I	L	R
N	H	M	R	C	U	A	L	M	A	R	P	A	S	L	M	I	P	O	U
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M	P	S	Z	T	S	R	I	U	G	G	A	S	Z	A	R	G	I	I	H
I	U	A	S	S	R	L	N	D	I	S	P	E	R	S	I	O	N	T	T
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S	C	P	R	T	T	A	R	Q	U	A	R	T	E	R	W	A	V	E	O
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N	L	N	N	O	I	S	E	N	O	I	T	A	Z	I	T	N	A	U	Q

**TRANSMISSION
WAVEGUIDE
QUARTERWAVE
ATTENUATION
IMPEDANCE**

**FEEDBACK
NYQUIST
ROUTHURWITZ
ROOTLOCUS
BODEPLOT**

**BIASING
CURRENTMIRROR
CONTROLLABILITY
STATESPACE
OPAMP**



FIND THE SUDOKU?

1 2 3
4
7



Useful Websites

Orbital Horizons: A rebranded lifestyle and innovation platform by space enthusiasts focusing on sustainable space living, research, and interplanetary technology. <https://orbitalhorizons.com/>

stellar Dynamics: An independent research site founded by aerospace engineers, covering the latest in space exploration, astrophysics, and rocket technology. <https://stellardynamics.org/>

Lunar City Official Websites: The Lunar Research Council launched four new portals for Lunar City Administration, Research & Development, Economic Expansion, and Space Security to enhance global collaboration. <https://lunarcity.gov/>

MBS Space Institute: A newly launched website offering insights into their advanced academic programs in space science, robotics, and satellite technology. <https://mbsspaceinstitute.com/>

Elite Aerospace Works: A platform showcasing Elite's innovations in spacecraft design, propulsion systems, and aviation technology. <https://eliteaerospaceworks.com/>

Cosmic Nannies: A fun initiative connecting scientists and families supporting childcare during long-duration research missions at space agencies. <https://cosmicnannies.org/>

AstroCare Careers: A job portal dedicated to opportunities in space medicine and astronaut health research, linking professionals with global institutions. <https://astrocarejobs.com/>

Rozas Research Lab: A newly launched platform sharing information about Rozas' work in space materials, propulsion systems, and microgravity experiments. <https://rozaslab.org/>

Moxey Space Systems: A company offering AI-driven solutions for satellite communication, orbital analytics, and mission networking. <https://moxeyspace.ai/>

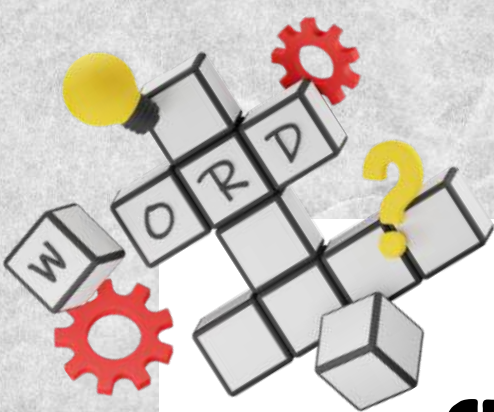
Americano Space Construction: A website showcasing Americano's expertise in constructing lunar habitats and modular spacecraft components. <https://americanospacebuild.com/>

Cravin' Cosmic Cuisine: An online platform providing space-approved nutritional meals for astronauts and researchers. <https://cosmiccuisine.net/>

A I B O T S

- Nano Banana – Google’s cutting-edge AI model for ultra-fast, high-quality image generation and editing with exceptional style and character consistency.
- Writesonic – Generate blog posts, social media captions, and creative content instantly.
- DALL·E 3 – Transform your imagination into stunning digital art in seconds.
- Beautiful.ai – AI-powered tool to create professional and visually appealing presentations.
- Aiva – Compose original music tracks for videos, games, or personal projects.
- Synthesia – Convert scripts into realistic AI-generated videos with avatars.
- Notion AI – Enhance your note-taking, summaries, and project planning with AI assistance.
- Runway – Advanced AI video editing and visual effects at your fingertips.
- NoiseZero – Remove background sounds, echo, and enhance audio clarity in real time.
- ContentBot – Plan, write, and schedule social media posts efficiently.
- Lavender AI – Craft high-converting cold emails with AI assistance.
- Tidio – Smart AI chatbots to automate customer support and website interactions.





CROSSWORD PUZZLES

ANSWERS





FIND THE WORD?

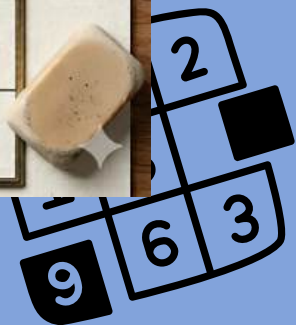
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THE SUDOKU

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8	4	2	9	5	6	3	7	 PEN
9	7	1	2	3	8	4	6	 VINTAGE PEN
5	6	3	1	7	4	2	8	9
2	9	4	8	6	6	5	3	1
6	5	8	3	2	1	9	7	4
3	1	7	4	9	5	6	2	8
4	2	9	7	1	3	8	5	 PRECISION ERASER
1	8	5	6	4	2	7	9	
7	3	6	5	8	9	1	2	





Poems

Trust and Love

I want to tell you something that's truly
matter...

The love of Radha and Krishna, so
pure and clear...

The trust of Rukmini and Krishna, so
deep and rare...

One story speaks of love, the
other of trust...

Without trust, love cannot exist...

Love means understanding, supporting each other
until the last breath...

Even when mistakes are made love will pull you
through...

Trust is the foundation upon which
love is built...

When you love someone, you see the best in them,
even when they don't see it themselves...

Love is not about finding someone perfect...
it's about learning to love someone's
unconditionally...

No matter what happens I will hold & Handle
the problems of our life...

But I need you to hold my hand until
my last breath...

By the way the biggest joke is.... the one who understands the
true meaning of love

they didn't get the chance to show that...

Like Krishna...

Burjukati Govardhan Reddy
9922005274

What kind of love.. ♦?

Love can arise any number of times..
if you ask me..One thing is enough..
if we know what kind of love it is?

Is it love born of attraction?

Love born of worship?

Love born of joy?

Love born of jealousy?

If we ask ourselves this questions?

If Love born of attraction does not last long..

But the love born of that worship
stays with us till we die..

born of happiness stops at friendship..

A love born of jealousy heralds
the downfall of the other...

The pure love means the love
that parents show to their children,
the love that devotees show to God,
the love that birds show to nature..

Burjukati Govardhan Reddy
9922005274



నాన్న..

ఈ పదం చిన్నదే కానీ ఈ పదం వెనుక ఉన్న కథలు ఎన్నో...
ఈ అంతులేని ఆకాశంలో చుక్కలను లెక్కించడం ఎంత కష్టమో,
అంత కన్నా కష్టం అయినది ఒక తండ్రి ప్రేమను వర్ణించడం.
ఆ అంతులేని ఆకాశాన్ని చూడటానికి సహాయపడే కళ్ల లాంటి నాన్నను మాత్రం
ఎవరు ఎప్పుడు గుర్తించరు... మేఘాల నడుమ దాగి ఉన్న
నీటి బిందువులాగా మన ప్రాణాలకు అర అడుగు దూరంలో...
నాన్నను మనం మన కళ్లతో చూసినా గుర్తించని నిజం..
తాను చీకటిలో ఉండి మమ్మల్ని వెలుతురు వైపు నడిపించే
ఒక అందమైన చందమామ.
ఎంతో మంది వల్ల అమ్మ ప్రేమను వరిస్తూ కావ్యాలు
వ్రాశారు కానీ తండ్రి తన పిల్లల కోసం చేసిన త్యాగం మరచారు...
తన జీవితంలో ఎన్నో కష్టాలు ఉన్నప్పటికీ అన్నింటినీ
దాటుకొని మన కోసం మాత్రం ఎప్పుడూ చెరగని చిరునవ్వుతో
కన్నీరు దాచుకుంటాడు... అందుకే ఏమో మనం ఏమీ
అడగక ముందేనే
ఇచ్చే నాన్నను కాదు అని ఏమీ ఇవ్వని ఆ దేవుని ఇంకా
ఏమో కావాలి అని కోరుకుంటాం...
అందుకే ఆ అంతులేని ఆకాశం కూడా చీకటే — వెలుగులాంటి నాన్న
లేకపోతే...

Burjukati Govardhan Reddy
9922005274



Story

“A MOTHER’S PROMISE”

Once upon a time, in the bustling city of Hyderabad, lived a small girl named Akshaya and her father, Ram. Akshaya had lost her mother during her childhood, leaving a void in her young heart. She would often ask her father about her mother, and each time, Ram would gently reply, Your mother is always with you, Akshaya. Whenever you need her, she will come to you. Akshaya would then question, Then why is she not here? I need to see her.



But Ram, with a heavy heart, had no answer.. One night, as Akshaya slept in her room, she had a vivid dream. In her dream, her mother appeared before her, and Akshaya tightly hugged her. It was a bonding moment that felt as calm and peaceful as seeing a full moon floating on the ocean...Her mother spoke softly, Hey Akshaya , why do you always ask about me? I am always with you, just not in a physical way. You are my soul, remember that... Akshaya has unexpectedly woken up...and thought what happen to her mother before died... ❏ Question remain silent. The next day, In the heart of Hyderabad, Ram worked tirelessly, balancing his demanding job and raising Akshaya with love and care. Their bond was unbreakable, and Rams world revolved around his daughter's happiness...

However, Akshaya carried a secret burden. She possessed a unique ability to see glimpses of the future in her dreams, but she kept it hidden from everyone, including her father. These visions, though often cryptic, revealed events that were yet to unfold, leaving Akshaya feeling isolated and fearful of the consequences of sharing her secret... One fateful night, Akshaya had a dream that shook her to the core. She saw her father, Ram, lying in a hospital bed, surrounded by beeping machines...? then she awakens suddenly, believing it to be a dream, and falls asleep once more... The next day, she went to school



During the afternoon lunch break, she got a call from her dad's office. She picked up the call. The manager said, you have to come to the Kim's hospital immediately. She asked why, Is my dad safe? Then call was disconnected.... She went to the hospital. The dream had actually happened. He was lying on the bed surrounded by doctors. She was controlling her tears with a heavy heart. She went close; her dad was not responding anymore. The doctors said, wait outside, we will call you after this. She was breathing heavily and sat outside.. After some time, the doctor came out. Akshaya ran towards the doctor

asking what happened. The doctor silently shook her head and left. She had no idea what the doctor said and went to her dad's room. Ram was lying on her bed without any consciousness. Then she realized the doctor's behavior... After realizing she was not accepting her fate, she called her dad and cried. Without her dad, she was nothing. She had many plans to do with her father when she grew up,

now the dreams were collapsed, and she had no idea about the future...

All of a sudden, the alarm rang, causing her to wake up. Written by BGR

Burjukati Govardhan Reddy
9922005274

AJAY AND THE BENCH BY THE LAKE

Ajay was an ordinary college student who often walked home instead of taking the bus. On his way, there was a quiet lake with an old wooden bench beneath a banyan tree.

He liked sitting there, even if just for five minutes. It wasn't special to anyone else, but to Ajay, it felt like the only place where the world slowed down. One evening, after a long and tiring day, Ajay sat on the bench, watching the ripples in the water when he noticed an old man struggling to pick up a dropped bag near the shore. Without hesitation, Ajay went and helped him. The old man smiled and said,

"You remind me of my grandson. He used to sit with me by a lake like this." They ended up talking for a while – about the city, about college, about life. For Ajay, it was just a small act of kindness, but for the old man, it filled a lonely evening with warmth. From then on, whenever Ajay passed the lake, he would check if the old man was there. Sometimes they spoke, sometimes they just sat in silence, watching the water.

Neither of them needed to say much – the company was enough. Years later, when Ajay had finished college and moved to another city,

he often thought about that bench by the lake. It reminded him that life isn't made up of big victories or dramatic moments, but of quiet connections that stay in the heart. Not every story needs an ending – some just leave a soft place in our memory, like the bench by Ajay's lake.



Ajaykumar M
9923005234

ECHOES OF THE LOST MELODY

Once upon a time, in the serene town of Mysuru, lived a boy named Arjun and his grandmother, Meera.

Arjun had lost his parents in a car accident when he was just five, and since then, Meera became his entire world. She often told him,

"Your parents are like music in the wind, Arjun. You can't see them, but if you listen carefully, you will always hear their melody."

One rainy night, as Arjun drifted into sleep, he dreamt of a vast concert hall filled with golden light.

In the center stood his mother, singing a lullaby he faintly remembered from childhood. His father played the flute beside her.

Arjun ran toward them, calling out, "Don't go. please stay."

But just as he was about to reach, the sound faded, and the stage dissolved into silence.

The next morning, he woke with tears in his eyes and a strange tune echoing in his mind.

That day, while walking to school, he passed by a music shop. From inside, he heard the same melody being played on a piano.

Startled, he entered, only to find no one inside except an old piano covered in dust. The keys had pressed themselves, leaving him in awe.



Burjukati Govardhan Reddy
9922005274

“THE FOOTPATH UMBRELLA”

Ramesh was an ordinary college student who walked home every evening because he could not afford the daily bus fare. On the way, there was a narrow footpath beside a noisy highway where a broken streetlight flickered above a small tea cart. He liked standing there for five minutes, even if he was tired, because that tiny patch of light made him feel safe in a world that moved too fast.



One rainy evening, after a long day of classes and part-time work, Ramesh stood under the broken light when he noticed an old woman trying to cover her vegetables with a torn plastic sheet. The rain was soaking everything. Without thinking, he opened his own umbrella and held it over her stall. The woman looked up at him and said, “My son used to stand like this for me. He left the city many years ago.” They spoke for a while about lost people, about how rain feels different when you are alone, about how children grow and disappear into life.

For Ramesh, it was only sharing an umbrella, but for the woman, it was as if someone had returned for a few minutes. After that day, whenever Ramesh passed that footpath, he would look for her stall. Sometimes they talked, sometimes he only held the umbrella while she worked in silence.

Years later, when Ramesh finally got a stable job and moved away, he often remembered that flickering streetlight and the old woman under the rain. It reminded him that the smallest kindness can become the warmest memory in someone else's cold life. But sometimes, on rainy evenings, he would remember that flickering streetlight, the quiet footpath, and the old woman protecting her vegetables from the rain.

And he would smile, realizing that the smallest kindness we offer someone might become the warmest memory they carry through the coldest days of their life.

Ajaykumar M
9923005234

THE WINDOW NEAR THE STAIRCASE

Lavanya was a quiet college student who always waited near the back staircase instead of rushing out with her classmates. Beside the stairs there was a tall dusty window that looked out over the railway tracks. She liked standing there for a few minutes after class, watching trains come and go, because it felt like the only place where her thoughts could breathe.

One evening, after failing an important test, she stood there wiping her tears when she noticed a security guard sitting on the steps, holding a photograph in his shaking hands. She did not ask anything, but she sat beside him. After a long silence, he said, "This is my daughter. She used to wave at trains from this window when she studied here." They spoke quietly about children who move away, about how empty places hurt more than crowded ones, about how some losses never get smaller.



That was when she noticed someone sitting on the steps nearby. It was the college security guard, a middle-aged man who usually greeted students at the gate with a polite nod. But today he looked different. His shoulders were bent forward, and in his hands he held an old photograph. Lavanya noticed his hands trembling slightly. She did not ask questions. Instead, she simply sat down beside him on the steps. They remained silent for several minutes, listening to the distant sound of a train approaching.

Abiraman S
9923005270



Pencil Art



~ Maram.Harshitha



~ Maram.Harshitha



~ Maram.Harshitha

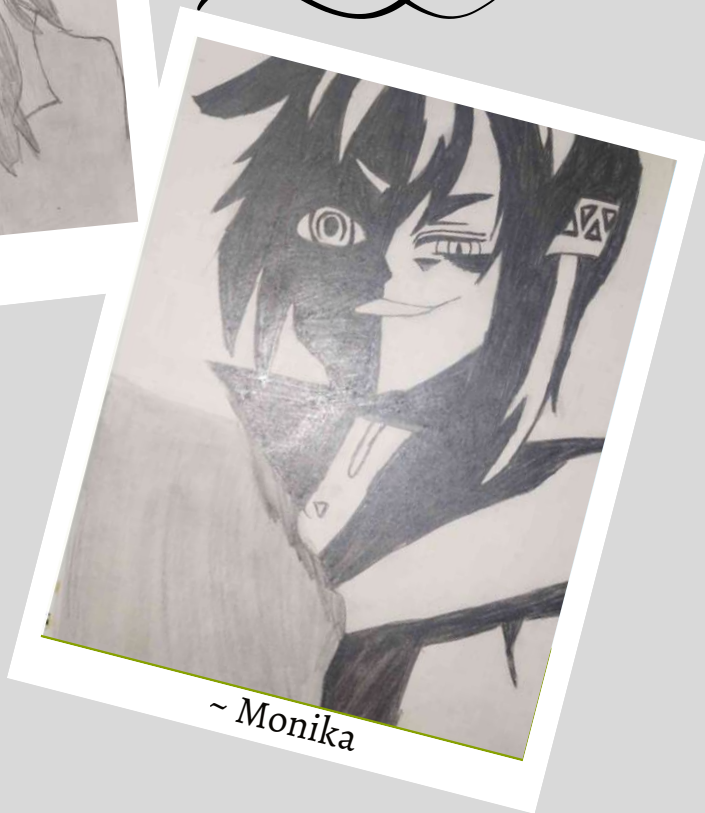




~Monika



~ Monika



~ Monika





~ Y Rama Krishna

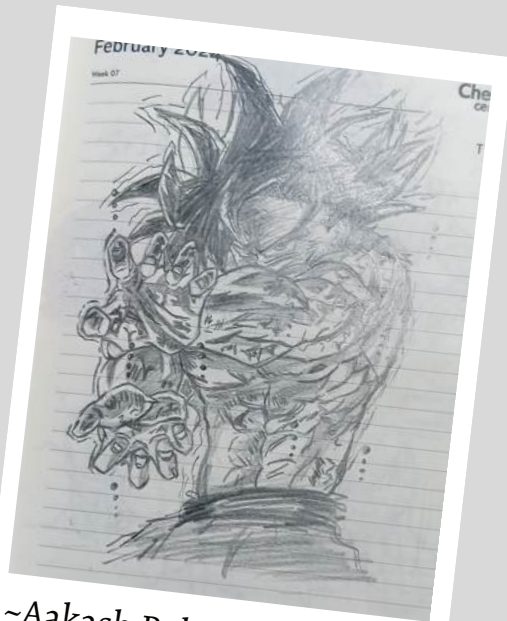


~ Aakash Bala Prajapathi S



~ Aakash Bala Prajapathi S





~Aakash Bala Prajapathi S



~ Aakash Bala Prajapathi S



~ Maram.Harshitha





Y Ramakrishna



Y Ramakrishna



Y Ramakrishna

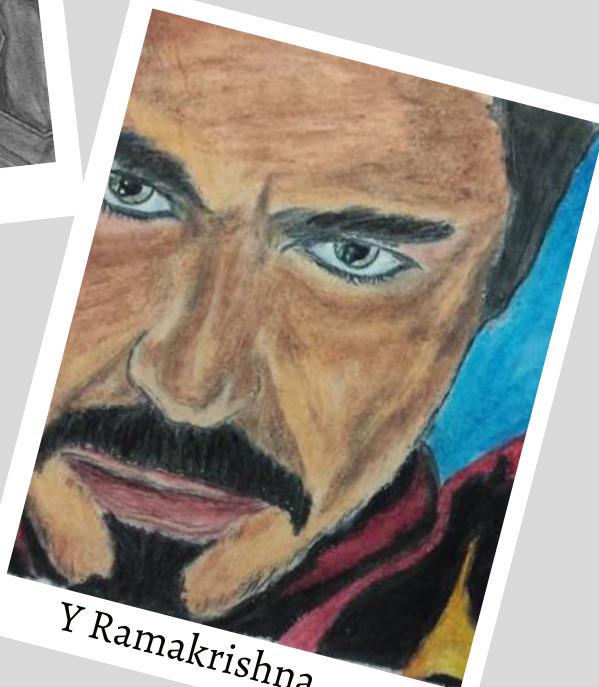




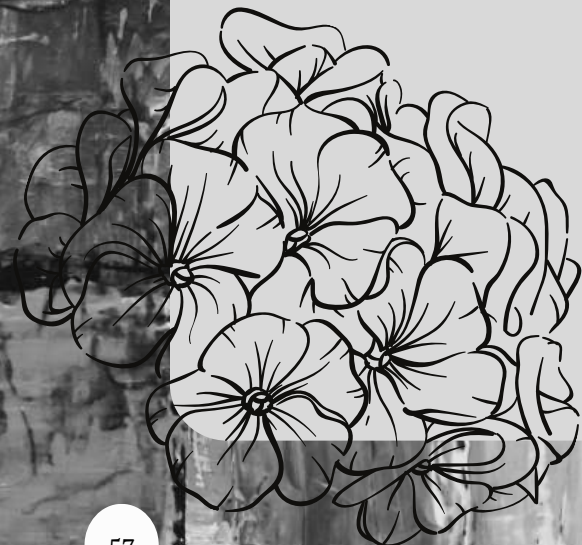
Y Ramakrishna



Y Ramakrishna



Y Ramakrishna

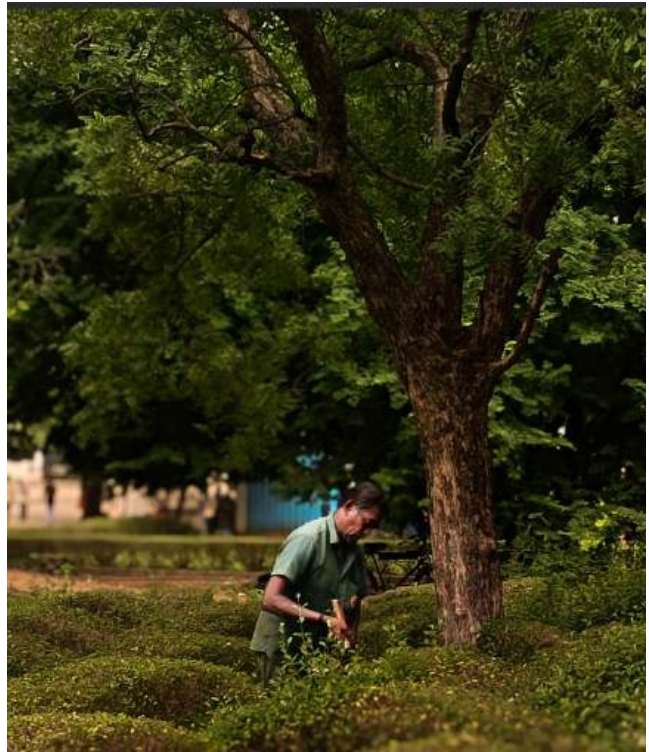




Photography



VENNIMUTHU(9923005267)



VENNIMUTHU(9923005267)



VENNIMUTHU(9923005267)



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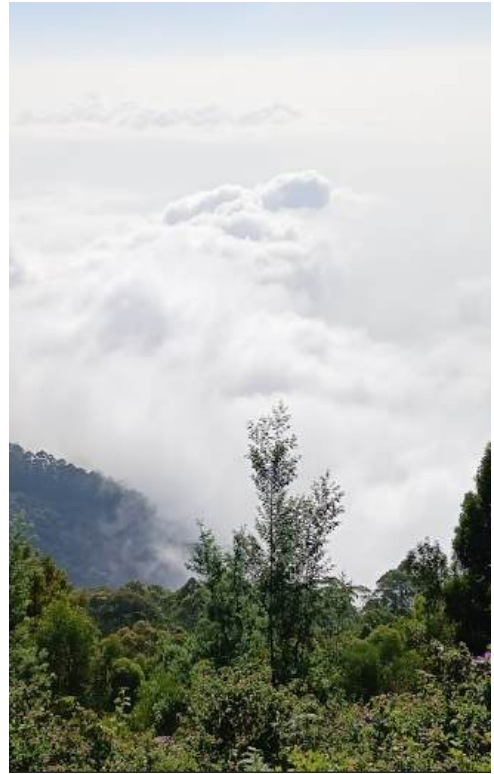
VENNIMUTHU(9923005267)



~ Aakash Bala Prajapathi S



~ Aakash Bala Prajapathi S



~ Aakash Bala Prajapathi S



YESWANTH REDDY



~ Aakash Bala Prajapathi S



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ABHINAV KRISHNA(9923005006)



~Syed Haroon



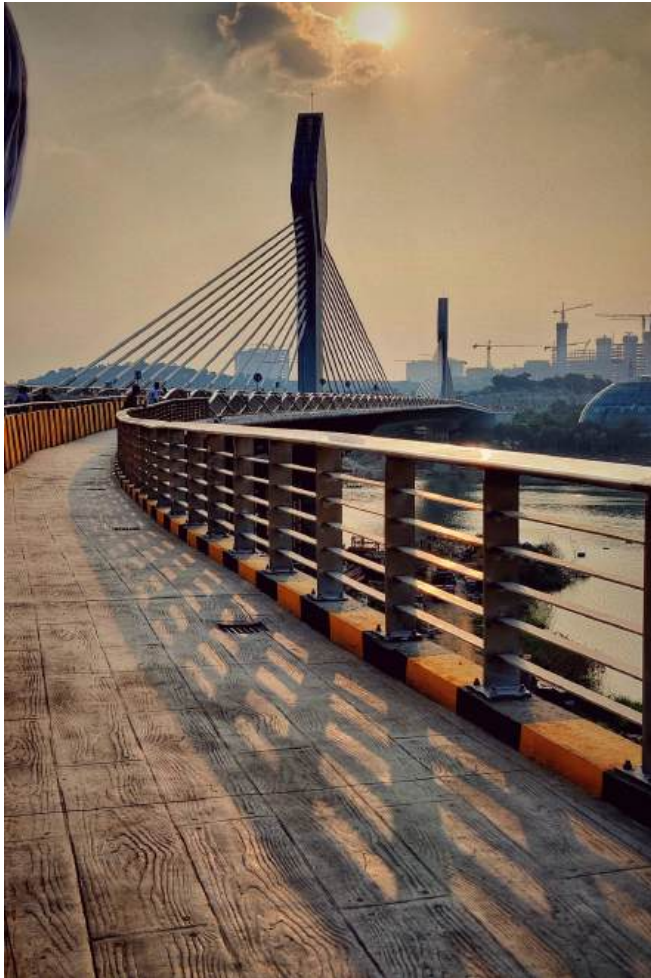
~Pavan Kumar



~Pavan Kumar



~Charan Babu



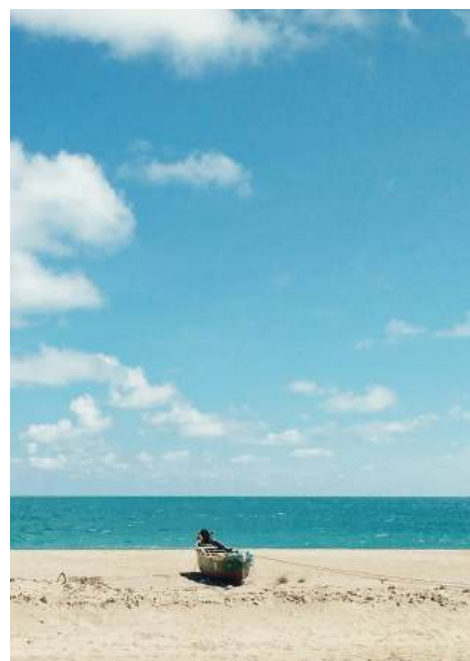
~Pavan Kumar



~Syed Haroon



~Karkuvel M



~Mohanaprasath

Feedback from students



Bhumireddy Himabindu
IV – ECE – E

I feel truly grateful to have studied in the ECE department. The faculty have been very inspiring throughout my journey. They not only taught us the concepts of communication, electronics etc.,... but also guided us in applying them to real-world problems. Their constant support and mentorship helped me develop confidence in both academics, projects and during my placements too. I am forever Thankful them for shaping my life more beautiful

I feel privileged to have my photo featured in the magazine by the ECE department for my placement. The guidance from faculty and the curriculum have helped shape my career path. I'm thankful for the opportunities and support I've received. Kudos to the ECE department for recognizing student achievements - it motivates us to strive further.



Thati Swathi
IV – ECE – A



Y. Bharath Kumar
III - ECE - F

The faculty is always available to support and encourage us as we engage in coursework, projects, and research. We acquire important knowledge and exposure as we take part in workshops, seminars, guest lectures, and technical fests. The faculty continuously inspires us as we participate in competitions, paper presentations, and innovation activities. We are urged to develop our leadership and collaboration abilities as we plan technical and cultural events. Under their direction, we gain technical proficiency, confidence, and inventiveness as we learn and mature.

Our Department has played a key role in my academic and professional development. The faculty members have been supportive, approachable, and have provided valuable guidance throughout the course. I would suggest enhancing industry exposure by conducting more workshops, internships, and training programs in core domains like VLSI, Embedded Systems, IoT, and AI/ML. Increasing collaborations with industries and improving placement opportunities in core companies would further strengthen the department. Overall, I am thankful to the department for its continuous efforts in shaping us as competent engineers and for providing a positive learning environment.



T. Komali
III- ECE - F



Sri Darshni Thayalan
III - ECE - F

My experience with the Electronics and Communication Engineering department has been truly rewarding. The faculty were always supportive and guided us in academics, projects, and research. I gained valuable knowledge through workshops, seminars, guest lectures, and technical fests. One of the highlights was the Robo Car Racing event by Aadhya Skills, where our team proudly secured 1st place. The department also encouraged us to take part in competitions and organize events, which helped me build confidence, teamwork, and technical skills.

Placements

Our Department has an abundance of job opportunities available to students in both core and software industries, with several renowned companies like genpact, neyes, zuci systems, comcast, accenture, zifo, foxconn, tcs and more are being potential employers.

OUR RECRUITER'S

amaDEUS

TATA ELECTRONICS

Capgemini



HEXAWARE



LTIMindtree

L&T Technology Services

tcs



Deloitte.

K P I T

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Delphi - TVS Technologies



SAVANTIS



mSPANd



COVALLENCE
Value Driven Excellence

Placements

aMADEUS -12.77 LPA



MOHAMED ANWAR M
IV - ECE - A

TATA ELECTRONICS -6 LPA



BHUMIREDDY
HIMABINDU
IV - ECE - E

FROM THE LIST OF 219 STUDENTS, A SIGNIFICANT NUMBER SECURED PLACEMENTS ACROSS REPUTED COMPANIES.

MAJOR RECRUITERS INCLUDE TRANSENERGY, DELPHI TVS, CAPG, LTTS, HEXAWARE, TCS (NINJA), WIPRO, NANOCHIP, AMADEUS, AVASOFT, LTI MINDTREE, KPIT, DELOITTE AND OTHERS. WE APPLAUD EVERY STUDENT FOR THEIR DEDICATION, PREPARATION, AND SUCCESS IN OBTAINING THESE ROLES.

GALLERY

Our department regularly organizes workshops, activities, and celebrations to ensure a well-rounded academic experience. This year, we proudly hosted Euphoria 2025, a vibrant departmental celebration showcasing student talent and innovation. We also conducted the MATLAB Chapter Coding Challenge, encouraging problem-solving and technical creativity among students. Additionally, we launched an Embedded Systems event, marking the beginning of an exciting journey into advanced hardware and software integration



EUPHORIA 2025 — WHERE CREATIVITY MEETS COLLABORATION, AND INNOVATION COMES ALIVE!



BLENDING TECHNOLOGY WITH IMAGINATION, WE CAME TOGETHER TO BUILD SOMETHING MEANINGFUL — A SMALL STEP TOWARD SHAPING THE FUTURE!



IT'S TIME TO CELEBRATE OUR CHAMPIONS! CONGRATULATIONS TO ALL THE WINNERS FOR THEIR AMAZING PERFORMANCES AND HARD WORK!

INTRAMURAL EVENTS

MATLAB CHAPTER - CODY CHALLENGE



WITH IMMENSE PLEASURE, WE WELCOME YOU ALL TO THE MATLAB CHAPTER - CODY CHALLENGE! LET THIS EVENT INSPIRE INNOVATION,



THIS CHALLENGE PUSHED US TO SOLVE COMPLEX PROBLEMS USING MATLAB, TESTING OUR CREATIVITY AND ANALYTICAL SKILLS.



THE WAIT IS OVER! LET'S APPLAUD THE BRILLIANT MINDS WHO SHOWCASED EXCEPTIONAL CODING TALENT AND EMERGED AS THE WINNERS OF THE CODY CHALLENGE 2025!"

LECTURE EVENTS EMBEDDED SYSTEM



WITH GREAT ENTHUSIASM, WE WELCOME EVERYONE TO THE GUEST LECTURE ON 'EMBEDDED SYSTEMS: THE POWER OF BITS IN ACTION' — AN INSPIRING SESSION EXPLORING THE BRILLIANCE OF TECHNOLOGY IN MOTION



THIS SESSION GAVE US DEEP INSIGHTS INTO THE WORLD OF EMBEDDED SYSTEMS, AUTOMOTIVE ECU VALIDATION, AND REAL-TIME INDUSTRY PRACTICES



WE EXTEND OUR HEARTFELT GRATITUDE TO MR. VIGNESH FOR HIS TIME, KNOWLEDGE, AND MOTIVATION

NAMBI NARAYANAN



"Truth, like science, always finds its way."